RIVER TILL RESTORATION STRATEGY

MONITORING PLAN

FINAL REPORT

Prepared for

Tweed Forum

Prepared by

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TABLE OF CONTENTS

1. Introduction 1
2. Baseline Data Collation and Review 4
3. Gap Analysis 6
4. Summary Site Details 7
   - 4.1 Site 1 - Wooler Water Bank Stabilisation 7
   - 4.2 Site 2 - Wooler Haugh Head 7
   - 4.3 Site 3 - River Glen Kirknewton Weir 7
   - 4.4 Site 4 - River Glen Reedsford Embankment Breach/ Planting 7
   - 4.5 Site 5 - River Glen Embankment Setback 7
   - 4.6 Site 6 - River Glen Akeld Riparian Fencing 8
   - 4.7 Site 7 - Lilburn Weir Removal 8
   - 4.8 Site 8 - Lilburn Riparian Fencing 8
   - 4.9 Site 9 - Till Fenton Riparian Fencing 8
   - 4.10 Site 10 - Breamish Bank Stabilisation 8
   - 4.11 Site 11 - Breamish Restoration (Harehall Hope to Bewick Bridge) 9
5. Monitoring Measures and Determination of Objectives 10
   - 5.1 Monitoring Measures 10
     - 5.1.1. Topographic Surveys 10
     - 5.1.2. Geomorphic Assessments/ Fluvial Audits 10
     - 5.1.3. Sediment Sampling 11
     - 5.1.4. Fixed Point Photography/ Drone Footage 11
     - 5.1.5. Ecological Surveys 11
     - 5.1.6. Hydrological Monitoring 11
   - 5.2 Determination of Site Specific Monitoring Objectives and Measures 12
6. Prioritised Work Packages 17
7. References 18

LIST OF FIGURES

Figure 1.1 Overview of site locations 3

LIST OF TABLES

Table 1.1 Key sites pre-selected for inclusion in the Monitoring Plan 1
Table 2.1 Existing data for sites of interest 4
Table 3.1 Data gaps identified for each site 6
Table 5.1 Monitoring objectives and measures 13
Table 6.1 Priority level of sites 17

LIST OF APPENDICES

Appendix A: Individual Site Work Packages

1. INTRODUCTION

The River Till is a gravel-bed river located in Northumberland with a catchment area of c.950km^2. It is a major tributary of the River Tweed and drains the Cheviot Hills as it flows east and then northwest towards its confluence with the Tweed near Coldstream. The River Till has five major tributaries which are the River Breamish (as the Till is known as it drains the Cheviot Hills), River Glen, Wooler Water, Hetton Burn and Lilburn Burn. The catchment supports a wide range of plant and wildlife species, which has led to it being designated as a Site of Special Scientific Interest (SSSI) and being part of the River Tweed Special Area of Conservation (SAC). Historical river and floodplain management has, however, detrimentally impacted the River Till and its tributaries and it has been classified by the Environment Agency as a Heavily Modified Water Body (HMWB) with seven Till SSSI units currently having a degraded physical and ecological condition and being categorised as not in 'favourable condition'.

To improve and restore the Till and its tributaries to a 'favourable condition' a River Till Restoration Strategy has been developed by Tweed Forum, the Environment Agency and Natural England. The aim of the River Till River Restoration Strategy (RRS) is to achieve the above by taking a catchment-scale approach that addresses the detrimental physical modifications contributing to the current status of the river. The RRS documents the visions of the strategy along with an assessment of the current physical condition of the catchment, possible methods to improve its physical condition (taking into account the potential constraints) and proposals on how to deliver the potential restoration measures outlined. To assess the success of these restoration measures in terms of improving the physical and ecological condition of the river environment of the catchment, pre- and post-construction monitoring of individual restoration sites, is required.

cbec was commissioned by Tweed Forum to finalise the Till Catchment Monitoring Strategy for 11 sites[1] throughout the catchment (a list of which is provided in Table 1.1), which have had, or plan to have, restoration works undertaken. For each restoration site a separate work package has been developed that sets out the specific monitoring objectives.

Table 1.1 Key sites pre-selected for inclusion in the Monitoring Plan

River Name	Individual Restoration Sites
Wooler Water	○ Wooler bank stabilisation ○ Haugh Head ford removal
River Glen/ Bowmont Water	○ Kirknewton weir removal ○ Reedsford embankment breech/ planting ○ River Glen embankment set back at Lanton/ Kirknewton/ College. ○ River Glen Akeld riparian fencing
Lilburn	○ Lilburn weir removal ○ Lilburn riparian fencing
Till main stem	○ Fenton riparian fencing
Breamish	○ Bank stabilisation/ grey bank protection replacement ○ Breamish restoration: Harehope Hall to Bewick Bridge

[1] Pre-selected by Tweed Forum.

To ensure the production of an effective and robust monitoring strategy, the PRAGMO (Practical River Restoration Appraisal Guidance for Monitoring Options) approach has been considered throughout the development of the work packages. The PRAGMO guidance document (Hammond, et al., 2011) provides guidelines on identifying suitable measures to monitor physical and ecological changes within a range of different restoration sites, through the identification of Specific, Measurable, Achievable, Realistic and Time-bound (SMART) objectives. Specifically, the SMART approach aims to focus the monitoring process through key questions such as:

- ***What is achievable?*** Determined through a review of what has been achieved at other, similar sites.
- ***What is realistic?*** What resources are available in terms of people/ capacity, budget and time that can be allocated to a particular site? Does data already exist and what can be collected?
- ***Over what timeframe?*** Are there any restrictions on timescales for delivery (e.g. funding deadlines, seasons, ecological windows, etc.)?

Through consideration of these fundamental points, a cohesive and achievable set of monitoring objectives and the required measures, can be identified.

This report documents the approach taken to determining monitoring objectives and priorities throughout the 11 pre-selected sites in the Till catchment. This includes an initial review of the sites and associated available datasets (**Section 2**) and analysis undertaken to identify gaps within the existing data (**Section 3**). Summary site details are provided in **Section 4**, and in **Section 5**, details of the monitoring objectives and measures are stated. Finally, through liaison with Tweed Forum, an assessment of priority sites was made, and the relevant work packages then produced (**Section 6**). The final work packages are provided in **Appendix A.**

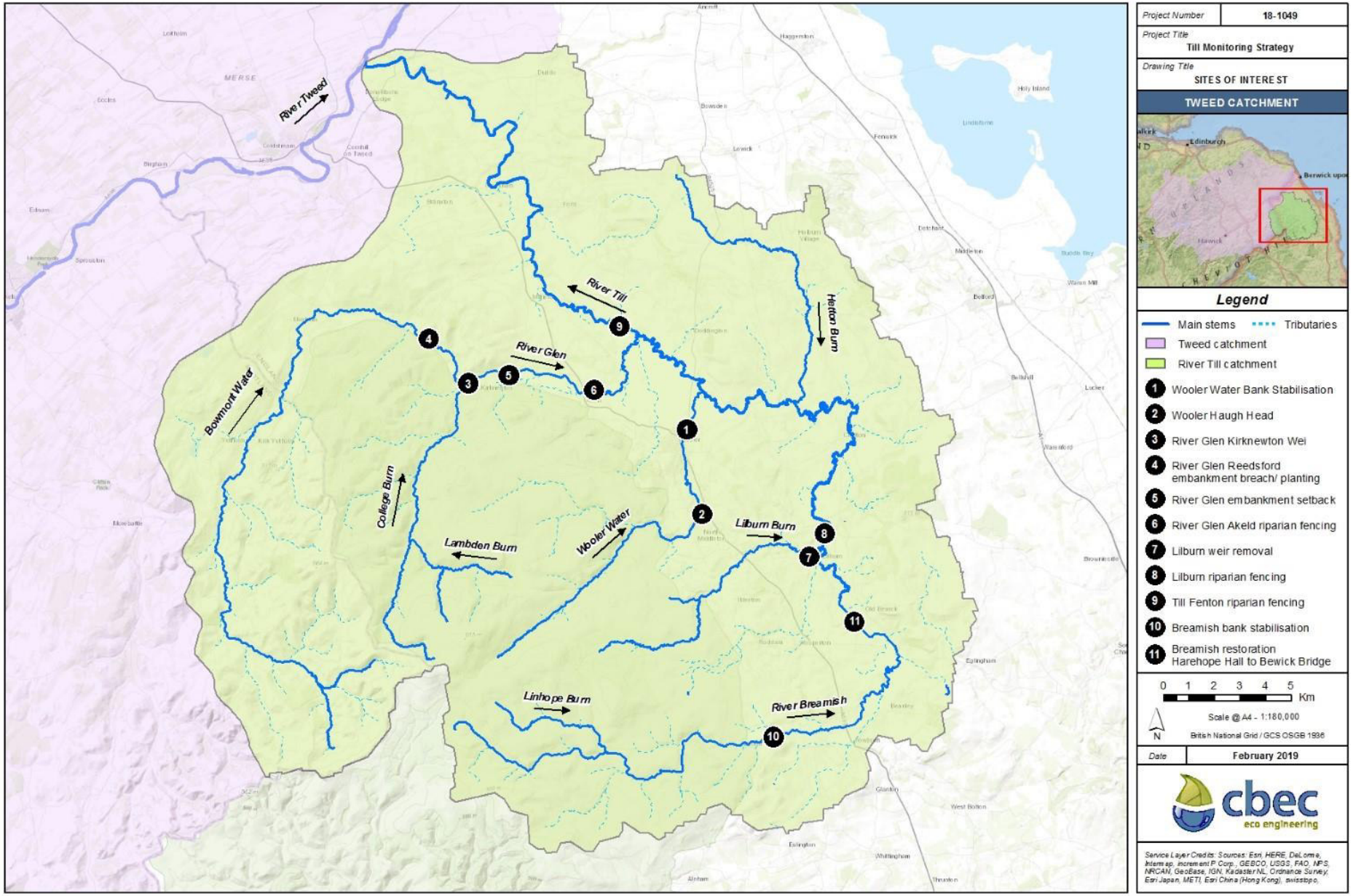


Figure 1.1 Overview of site locations

2. BASELINE DATA COLLATION AND REVIEW

Comprehensive baseline data collected over a suitable time period will be key to the success of the monitoring exercise. In liaison with Tweed Forum and the Environment Agency, a list of all data available for each of the eleven sites, has been collated. The table below provides a summary of this data.

Site 6 (Akeld riparian fencing) is greyed out, as whilst this formed part of the original scope, Tweed Forum advised during the course of the project that this site was to be descoped due to the minimal extent and nature of the works.

Table 2.1 Existing data for sites of interest

ID	Site	Data/ Source
1	Wooler Water bank stabilisation	Photos – Pre, during and post construction (cbec) Photos – Pre construction (Dickinson/ Tweed Forum) Pre-construction topographic data (cbec) Pre-construction modelling (cbec) Drawing files (cbec) Project report (cbec) Post-construction topographic data (cbec) Monitoring Report (cbec)
2	Wooler Haugh Head ford removal	Photos (cbec) Wider Wooler Water photos (Tweed Forum) Wider Wooler Water drone footage (Tweed Forum/ Rutherfords) Wider Wooler Water consenting application documentation for 'Way to Wooler' site (Tweed Forum/ Rutherfords) Topographic data (cbec) Project report (cbec) Flood mapping study report (JBA) Wooler Water geomorphology study (Jacobs) Drawing files (AECOM) Sediment transport modelling report (cbec) Still frames from timelapse footage (provided by EA) Ortho-rectified drone footage (EA) Draft design drawings (AECOM) Peer review (AECOM) Flood model and topographic data (JBA/ Storm Geomatics) *Ecological survey report - pending*
3	River Glen Kirknewton Weir removal	Hydromorphology/ options appraisal report (AECOM) Options development report and appendices (including options design drawings) (Tweed Forum) Timelapse video (available online at https://www.youtube.com/watch?v=JyTv4_MMV8k) Before and After photo (EA).
4	River Glen Reedsford embankment breach/ planting	Report/ case study sheet (Tweed Forum) Photos (Tweed Forum)

ID	Site	Data/ Source
5	River Glen embankment setback	River Glen DTM dated 25th April 2018 (AQUAUoS) River Glen DTM dated 1st November 2018 (AQUAUoS) River and floodplain management on the River Glen – Options report (AECOM) Ecological assessment for Lanton (EA) Draft design/ model report (AQUAUoS) Concept drawings (AQUAUoS) Embankment works volumes (AQUAUoS) Summary model results for indicative design (AQUAUoS) Drone imagery **still shots only** (Tweed Forum and EA) Lanton Floodbank Removal Plan (Tweed Forum)
6	River Glen Akeld riparian fencing	No data available
7	Lilburn weir removal	Before and after photos (in pdf) Small number of photos in jpeg format (supplied by EA) Lilburn Weir removal_ Photo log with comments (supplied by EA) Pre-construction photo record (Tweed Forum) Options appraisal report (Envirocentre)
8	Lilburn riparian fencing	Lilburn riparian fencing plan (Tweed Forum) Photos pre-fencing (Tweed Forum)
9	Till Fenton riparian fencing	Report/ case study (Tweed Forum) Planviews of fenceline (Tweed Forum/ Simon Henderson) Historic channel change data (Tweed Forum/ Simon Henderson) Site photos pre-fencing (Tweed Forum)
10	Breamish bank stabilisation	Options report (cbec) Map (provided by EA) Photos (cbec) Photos (Tweed Forum) Log works description (Tweed Forum)
11	Breamish restoration: Harehope Hall to Bewick Bridge	Fluvial audit data (cbec) Breamish options appraisal report (cbec) Georeferenced photographs (cbec)

3. GAP ANALYSIS

Following the collation of the data available, an analysis was undertaken (using both information provided by Tweed Forum, and expert judgement) to identify definite and potential gaps within the existing datasets (Table 3.1). This information was important to consider during development of the Monitoring Plan, directly impacting on the potential monitoring work required/ possible at each site. The boxes shaded in grey represent sites which were part of the original part of the restoration strategy, but have now been descoped.

Table 3.1 Data gaps identified for each site

ID	Site	Data/ Source
1	Wooler Water bank stabilisation	No gaps identified
2	Wooler Haugh Head ford removal	**Flood Risk Assessment report** **No raw drone footage has been made available**
3	River Glen Kirknewton Weir removal	**No raw fluvial audit or geomorphic data**
4	River Glen Reedsford embankment breach and planting	**No extent of planting boundary has been obtained**
5	River Glen embankment setback	**No raw drone footage has been made available**
6	River Glen Akeld riparian fencing	**No data has been obtained for this site**
7	Lilburn weir removal	**All original photo files available from the EA.**
8	Lilburn riparian fencing	No gaps identified
9	Till Fenton riparian fencing	No gaps identified
10	Breamish bank stabilisation	No gaps identified
11	Breamish restoration: Harehope Hall to Bewick Bridge	**No topographic/ DEM data is available for this site**

4. SUMMARY SITE DETAILS

A summary for each of the sites included within the Till Catchment Monitoring Strategy is provided below. This details the location of the site within the catchment along with the work that has been undertaken to date or is to be undertaken at each site.

4.1 SITE 1 - WOOLER WATER BANK STABILISATION

The Wooler Water is a highly dynamic gravel-bed river and large flood events in 2008 and 2009 resulted in extensive channel adjustment to a section of the river immediately downstream of Wooler town. At the site in question, the river avulsed into a new location, resulting in the active erosion of a 140 m section of bank, 2-3 m in height that posed a risk to a nearby buried pipeline.

Design and construction work was commissioned at the site between 2014 and 2015. The eroding bank was reprofiled to provide a more stable angle and protected along the length of its base with a matrix of natural large wood material.

A next phase of monitoring is now ongoing, with annual (or post-flood) resurveys/ analysis of the site continuing. However, following discussions between Tweed Forum and project partners this site has been descoped.

4.2 SITE 2 - WOOLER HAUGH HEAD

The Haugh Head ford, located on the Wooler Water upstream of Wooler, Northumberland, is a significant barrier to fish passage. A number of assessments have been commissioned by Tweed Forum at the site, to assess options for mitigation of the structure. A large number of stakeholders and landowners have been involved in the discussions to date. Sediment transport and flood risk modelling studies have been undertaken throughout the immediate reach and wider Wooler Water extents, the results being used to inform the production of designs for the site.

4.3 SITE 3 - RIVER GLEN KIRKNEWTON WEIR

The Kirknewton weir was a defunct gauging structure, situated on the River Glen near Kirknewton, at OS NGR NT 91870 31026. Tweed Forum commissioned an options appraisal of the site in 2015, following which, outline design drawings were developed for each of the options identified. Full removal of the weir and associated structures was ultimately recommended, and construction has since been completed at the site to remove the weir. Following discussions between Tweed Forum and project partners this site has been descoped.

4.4 SITE 4 - RIVER GLEN REEDSFORD EMBANKMENT BREACH/ PLANTING

Tweed Forum have commissioned works at Reedsford, Northumberland (NT 89060 32515), with the aim of improving Natural Flood Management within the River Glen system, as well as helping to improve biodiversity at the site. The objective was to achieve this through implementation of a series of measures, including enlargement of an existing embankment breach (to allow greater channel – floodplain connectivity), rerouting/ erection of new fencing, and riparian planting.

4.5 SITE 5 - RIVER GLEN EMBANKMENT SETBACK

The River Glen is a high energy gravel bed system, which is constrained within a single thread channel. The surrounding floodplain has evidence of multiple palaeo-channels illustrating the historic dynamic nature of the river. Complete or partial embankment removal has been proposed to reconnect the

river to its surrounding floodplain from College Burn through to the A697 road crossing involving collaboration of several landowners.

4.6 SITE 6 - RIVER GLEN AKELD RIPARIAN FENCING

No information was obtained for this site during an initial search. Tweed Forum have since advised that at this stage, in terms of monitoring, no further consideration is required for this site.

4.7 SITE 7 - LILBURN WEIR REMOVAL

The Lilburn weir structure was initially identified for removal as part of the 'assisted natural recovery' approach of the Till River Restoration Strategy, with the aim of improving in-channel natural geomorphic process, as well as benefiting fish passage.

The structure, located upstream of the Lilburn confluence with the River Till at grid reference NU 04674 23735, has since been removed, with the channel and surrounding banks being reprofiled/ regraded in places.

4.8 SITE 8 - LILBURN RIPARIAN FENCING

The Lilburn Riparian Fencing scheme focusses on a dynamic section of the Upper River Till, between Netwon Mill and Newton Bridge, nr. Lilburn, Northumberland. The aim of the works was to install fencing along the left bank of the river to create a suitable riparian buffer zone, protected from livestock poaching. The fencing was located at such a distance as to reduce risk of damage due to future lateral channel migration.

4.9 SITE 9 - TILL FENTON RIPARIAN FENCING

This project has involved the installation of ~850 m of temporary fencing along a river bank at West Fenton, Northumberland. Riverbank erosion at the site was being exacerbated by livestock poaching. The fencing has been installed with the aim of protecting the immediate bank area from livestock, thereby encouraging vegetation to re-establish, improving stability along the bank.

4.10 SITE 10 - BREAMISH BANK STABILISATION

Options appraisal and outline design work has recently been undertaken at a highly dynamic reach of the River Breamish, located between Ingram and Brandon, Northumberland. The aim was to identify the most suitable restoration options to improve the physical condition of the river and protect vital infrastructure from future damage. In early 2018, following a detailed geomorphic assessment, a suite of options were identified, with two options being identified as potentially feasible. These were:

1. Sustainable bank protection measures in four key sections of erosion.
2. In the longer term, the re-routing of the access road to Ingram, diverting the route from it's current alignment within the active left hand floodplain.

Outline designs and visualisations for option 1 have since been developed and were submitted to Tweed Forum for consideration, in December 2018.

4.11 SITE 11 - BREAMISH RESTORATION (HAREHALL HOPE TO BEWICK BRIDGE)

The River Breamish, a tributary of the River Till, is a wandering gravel-bed river. Significant historical channel realignment and embankments have resulted in the physical condition within this reach being severely degraded. In winter 2017, cbec undertook a scoping study to identify potential options to improve the physical condition of the River Breamish between Harehope Hall and Bewick Bridge. Options included the setting back of embankments and channel realignment to a more natural planform.

Following a period of initial landowner consultation, Tweed Forum are now looking to progress baseline monitoring at the site, to determine the most suitable option to take forward to detailed design.

5. MONITORING MEASURES AND DETERMINATION OF OBJECTIVES

To monitor the success of the restoration measures implemented as part of the Till Restoration Strategy a set of specific, measurable, achievable, realistic, and time- bound (SMART) monitoring objectives were developed for each site. Careful consideration was given to ensure that the objectives set for each site were realistic and tailored to the ultimate aims and scale of the individual sites.

To achieve this, it was important to firstly understand the types of methods available for monitoring. Descriptions of those which form part of the final measures identified for the Till catchment sites, are provided in Section 5.1, alongside key considerations for each.

5.1 MONITORING MEASURES

5.1.1. Topographic Surveys

Pre and post-construction topographic surveys allow for detailed monitoring of how individual sites evolve and adjust as a result of restoration works. These detailed assessments can also be used to inform hydrological, hydraulic and morphodynamic (i.e. sediment transport) modelling of post-restoration channel evolution.

It's vitally important that the method used for surveying is considered if such monitoring is to be undertaken. The level of detail captured during any baseline survey, is required to be replicated for any repeat surveys, to ensure that the resulting surfaces are comparable.

5.1.2. Geomorphic Assessments/ Fluvial Audits

Geomorphic assessments/ fluvial audits are important tools for monitoring restoration works, as the data collected can help build an understanding of multiple criteria used to assess the success of implemented works. Such criteria include:

- geomorphic processes within the reach
- the related distribution of meso-habitats within the restoration reach
- channel evolution over time
- bed sediment distribution within meso-scale habitat units, to feed into analysis of ecological responses.

One benefit of employing this type of assessment is that data can be tailored to meet the needs of each individual project, i.e. the level of detail and type of information collected during each survey can reflect the nature of the site and the objectives of the work.

For example, for projects looking to assess/ monitor sediment regime, important characteristics and features to record might include reach typology, substrate, sediment inputs (i.e. tributaries, bank erosion), sediment storage (i.e. alluvial barforms), structures (particularly those which have potential impact on sediment regime such as weirs or flow deflectors) and in-channel large wood. For projects focussing on habitat, key information will include morphological units, sediment storage features, large wood, macrophyte presence, riparian vegetation cover and morphological pressures such as channel realignment and in-channel structures.

That said, it is recommended that any initial, baseline assessment is as detailed as possible, so that a full suite of pre-construction data is available for future comparative purposes. This minimises risk of issue if project objectives change or are adapted throughout the work.

5.1.3. Sediment Sampling

In conjunction with geomorphic assessments, sediment sampling can be undertaken throughout a reach/ multiple reaches, to monitor changes in sediment particle size distribution across not only the restoration reach but also within meso-scale habitat units. This allows us to understand channel evolution post-restoration, can be used to inform ecological responses to works, and can also be used to inform hydraulic/ morphodynamic modelling.

Key to this method of monitoring is to ensure that samples are collected from the same locations during each visit, to allow for direct comparison.

5.1.4. Fixed Point Photography/ Drone Footage

Repeat fixed point photography is a suitable method of qualitatively monitoring evolution of a site over time. Being a lower cost option in comparison to other monitoring approaches, fixed point photography and the collection of drone footage can often be undertaken at more regular intervals, allowing for a visual assessment of shorter-term changes/ channel adjustment.

One of the key considerations for fixed point photography is to ensure that key locations are pre-defined, so that photographs are directly comparable between visits. Drone footage should cover the same extents between surveys, again for the purpose of comparison.

5.1.5. Ecological Surveys

Ecological surveys allow for the collection of up-to-date data on presence of key species within in-channel, bankside and riparian/ floodplain areas. Such surveys can focus on a range of different species, depending on the focus of the project and the species known to inhabit a site. Often, focus is on protected species (e.g. bats, otters, water vole, pearl mussels) and/ or Invasive Non Native Species (INNS) such as Giant Hogweed, Japanese Knotweed or Himalayan Balsam.

The ecological response of a site to restoration works can be determined through collection of such data, making it possible to quantitatively assess increased presence of particular species, as well as providing an indication of any changes to habitat provision.

5.1.6. Hydrological Monitoring

Monitoring of site hydrology can be an important criteria for determining localised changes to flow regime as a result of implemented restoration works. This is a particularly useful approach to monitoring where a focus of the work is on water levels/ flooding, or where there are specific concerns from involved parties as to the impact of restoration works on river levels within a reach.

Hydrological monitoring can be undertaken through the installation of a temporary gauge, appropriately secured into the bed/ bank of the river channel. This allows for the collection of regular (i.e. hourly, daily, etc) water levels, over a specified length of time[2]. Regular visits are recommended to such sites, to avoid large volumes of data loss in the event that the instrumentation fails or is damaged. Other methods include visual assessment of stage boards.

[2] The length of time over which data can be collected can be dictated by the gauge instrument used.

5.2 DETERMINATION OF SITE SPECIFIC MONITORING OBJECTIVES AND MEASURES

For the larger scale sites involving extensive restoration such as bank reprofiling, embankment removal and/ or channel realignment, the recommended monitoring objectives are more comprehensive in order to cover a larger range of criteria. For the smaller sites, such as those involving installation of new fencing only, the monitoring objectives reflect the less extensive nature of the works and the requirement for a smaller range of criteria to be monitored.

The more complex monitoring objectives involve the gathering and interpretation of multiple quantitative datasets whilst the lighter touch monitoring objectives focus on collection of more basic project data and visual interpretation. Delivery timescales and budget constraints were also considered.

Table 5.1 provides a breakdown of each of the sites and the recommended monitoring objectives, in line with the PRAGMO approach. Detailed for each site are the measurable criteria (e.g. sediment size, topography), the date collection methods (i.e. types of survey) used to fulfil the monitoring objectives, as well as a brief description of the initial project targets and suggested monitoring frequency. Clearly defining the targets for each of the sites ensures the proposed monitoring is both cost effective and aligned to the project targets.

Table 5.1 Monitoring objectives and measures

Restoration Site	Initial Project Target	Construction Complete	Monitoring Objectives	Measureable	Method	Frequency
Wooler Bank Stabilisation	•Improve stability of bank/ resilience of bank to significant flood events. •Bank erosion along a stretch of the river was posing a threat to a floodbank, sewer pipe infrastructure and adjacent grazing land.	Yes	• To determine post restoration geomorphic adjustment (including flow paths, sediment distribution and erosion patterns).	Topography of site.	Detailed topographic surveys of river corridor	Pre works, post works and then biannually (or after significant flood events)
				Condition of implemented measures including signs of further adjustment/ hotspots for future risk.	Geomorphic assessment (i.e. fluvial audit) to quantify any change.	Bi-annually or after a significant flood event (whichever sooner) for at least at least 10 years.
Haughhead ford removal	•Assessment of feasibility (and impacts) of removal of ford structure at Haugh Head. •Ford currently unstable and is contributing to significant disruption of sediment transport processes, formation of bedforms/ habitat, and fish passage. Removal would aim to improve such longitudinal process.	No	• To determine the post restoration geomorphic adjustments; • To determine changes to sediment distribution as a result of the works; • To determine the physical habitat diversity of the restored reach. • The ecological response within restored and adjacent reaches, to document how the biota adjust to the changing physical habitat.	Topography of site.	Detailed topographic survey of river corridor	Pre works, post works and then biannually (or after significant flood events)
				In channel particle sizes.	Sediment sampling/ particle size analysis of riverbed sediment (wetted channel and sediment deposition) downstream of works.	Pre works, post works and then biannually (or after significant flood events)
				Presence of habitat/ species.	Ecological survey	Pre works, post works and seasonally for up to 5 years (or after significant flood events)
Kirknewton Weir removal	•Removal of redundant weir structure •Why? To improve sediment transport regime.	Yes	• To determine the post restoration geomorphic adjustments; • To determine the physical habitat diversity of the restored reach.	Visual condition of the site (including formation of bedforms and general bank condition).	Fixed point photography mapped with GPS co-ordinates for future use.	Seasonally (minimum twice a year) at least 10 years (or after significant flood events).
				Condition of the site.	Geomorphic assessment (i.e. fluvial audit) to quantify change.	Bi-annually or after a significant flood event (whichever sooner) for at least 10 years
				Topography of site.	Detailed topographic survey of river corridor.	Pre works, post works and then biannually (or after significant flood events)

Restoration Site	Initial Project Target	Construction Complete	Monitoring Objectives	Measureable	Method	Frequency
Reedsford embankment breech/ planting	•Enlarging of an embankment breach relocating fencing as a result of the works, and enhancing woodland and wetland vegetation •To increase flood water storage and improve habitat.	No	• To determine rate of establishment of riparian and wetland vegetation; • To determine the ecological response within restored and adjacent reaches, to document how the biota adjust to the improved riparian and wetland habitat.	Visual condition of the site including processes occurring and establishment of vegetation.	Fixed point photography mapped with GPS co-ordinates for future use.	Seasonally (minimum twice a year) at least 10 years (or after significant flood events).
				Presence of biota throughout the site.	Ecological survey	Pre works, post works and seasonally for up to 5 years (or after significant flood events)
River Glen embankment set back at Lanton/ Kirknewton/ College	•Assess and implement potential integrated management options across the floodplain from College Burn through to the A697 road crossing involving several landowners •To protect assets and utilities throughout the River Glen and College Burn floodplain from continued avulsion	No	• To determine the post restoration geomorphic adjustments; • To determine changes in hydrology and hydraulics post-works of restored and adjacent reaches, to identify the impact on flood levels.	Visual planform/ condition of the site.	Georeferenced Drone Survey	Pre works, immediately post works and annually (or after significant flood events, whichever sooner).
				Topography of site.	Detailed topographic surveys of river corridor	Pre works, immediately post works and annually (or after significant flood events, whichever sooner).
				Site hydrology.	Temporary hydrological data logger(s)/ gauge(s)	Fixed in place prior to works and remaining in place for 10 years post-works.
				Condition of implemented structures/ measures and effects on/ impacts to wider reach as result of the works including signs of further adjustment/ hotspots for future risk.	Geomorphic assessment (i.e. fluvial audit) to allow quantification of any post-works adjustment.	Bi-annually or after a significant flood event (whichever sooner) for at least 10 years

Restoration Site	Initial Project Target	Construction Complete	Monitoring Objectives	Measureable	Method	Frequency
Lilburn weir removal	•Removal of redundant weir from the Lilburn Burn •To restore natural geomorphological and hydrological processes, with associated benefits to habitat and fish passage.	Yes	• To determine the post restoration geomorphic adjustments; • To determine the physical habitat diversity of the restored reach	Visual condition of the site (including formation of bedforms and general bank condition).	Fixed point photography mapped with GPS co-ordinates for future use.	Seasonally (minimum twice a year) at least 10 years.
				Condition of the site (including formation of bedforms and general bank condition).	Geomorphic assessment (i.e. fluvial audit) to allow quantification of any post-works adjustment.	Bi-annually for a least 10 years. **No post works data but repeated surveying will provide information on the rate of channel adjustment.*
Lilburn riparian fencing	•Installation of new fencing along a reach of Lilburn Burn •To improve resilience/ decrease risk of damage from lateral channel migration.	Yes	• To determine whether future adjustment at the site poses further risk to installed fencing or other adjacent areas of bank.	Visual condition of fencing and visual inspection of the river in this location.	Fixed point photography mapped with GPS co-ordinates for future use.	Seasonally (minimum twice a year) at least 10 years.
Fenton riparian fencing	•To install temporary fencing along a stretch of actively eroding river bank on the River Till Mainstem •To protect from livestock access, to allow the riverbank the opportunity to stabilise and re-vegetate.	Yes	• To determine whether future adjustment at the site poses further risk to installed fencing or other adjacent areas of bank.	Visual condition of fencing and visual inspection of the river in this location.	Fixed point photography mapped with GPS co-ordinates for future use.	Seasonally (minimum twice a year) at least 10 years.

Restoration Site	Initial Project Target	Construction Complete	Monitoring Objectives	Measureable	Method	Frequency
Breamish bank stabilisation/ grey bank protection replacement	•To install sustainable bank protection measures along four sites of the River Breamish •To protect a vital access road from damage due to lateral channel migration.	No	• To assess stability of the implemented bank stabilisation measures; • To monitor other areas of the reach to ensure no further adjustment poses risk to the access road.	Visual condition of the implemented structures/ measures and general condition of the wider reach.	Fixed point photography mapped with GPS co-ordinates for future use.	Seasonally (minimum twice a year) at least 10 years (or after significant flood events).
				Condition of implemented structures/ measures and effects on/ impacts to wider reach as result of the works including signs of further adjustment/ hotspots for future risk.	Geomorphic assessment (i.e. fluvial audit) to assess wider channel change.	Within first year of installation, or after first major flood event (whichever sooner).
Breamish Restoration: Harehope Hall to Bewick	• To improve the physical and ecological of this reach of the River Breamish. •Improvement floodplain connection.	No	• To determine the post restoration geomorphic adjustments; • To determine the physical habitat diversity of the restored reach. • To determine changes in hydrology and hydraulics post-works of restored and adjacent reaches, to identify the impact on flood levels.	Visual condition of the site (including formation of bedforms and general bank condition).	Fixed point photography mapped with GPS co-ordinates for future use.	Seasonally (minimum twice a year) at least 10 years (or after significant flood events).
				Condition of the site (including formation of bedforms and general bank condition).	Geomorphic assessment (i.e. fluvial audit) to assess wider channel change.	Within first year of installation, or after first major flood event (whichever sooner).
				Visual planform/ condition of the site.	Georeferenced Drone Survey or topographical survey.	Pre works, immediately post works and annually (or after significant flood events, whichever sooner).
				Site hydrology.	Temporary hydrological data logger(s)/ gauge(s)	Fixed in place prior to works and remaining in place for 10 years post-works.

6. PRIORITISED WORK PACKAGES

The PRAGMO approach was used to develop a work package for each of the 11 pre-selected sites within the Till catchment. Each has been produced as a standalone document and all 11 are provided in **Appendix A**.

The intention is for each of the work packages to be a working document which can be referred to and updated as required. Each document provides a summary of the site and outlines the data currently available for the site along with a breakdown of the monitoring objectives, methods to be used, and recommended frequency/ timescales for each type of assessment recommended. In line with the PRAGMO approach the monitoring measures recommended aim to provide data which will clearly demonstrate if the projects objectives have been met.

To assist the TRRS Steering Group in prioritising the work between the sites, the work packages have been colour coded to show their priority. Prioritisation has been determined based on criteria such as the extents and nature of the restoration implemented or proposed, the extent of the monitoring requirements, and the status of each site (i.e. pre- or post-construction).

Sites have been separated into high, medium and low priority with sites colour coded red, orange and yellow, respectively. Table 6.1 lists the sites grouped into each category. With this information, the TRRS Steering Group can assess the individual sites within each category and determine the final order of progression based on feasibility (in terms of funding, programme, buy-in, etc).

Table 6.1 Priority level of sites

Priority Level	Site #	Site Name
Highest	2	Haugh Head ford removal
	5	River Glen embankment set back at Lanton/ Kirknewton/ College
	10	Breamish bank stabilisation/ grey bank protection replacement
	11	Breamish restoration: Harehope Hall to Bewick
Medium	4	Reedsford embankment breech/ planting
	7	Lilburn weir removal
Low	1	Wooler bank stabilisation
	3	Kirknewton weir removal
	8	Lilburn riparian fencing
	9	Fenton riparian fencing
Descoped	6	River Glen riparian planting

Additionally, within the work packages of each of the four sites still to be constructed (specifically, Sites 2, 5, 10 and 11), the assessment methods required in order to achieve the monitoring objectives for each site, have also been prioritised, to highlight the highest priorities in terms of baseline data collection.

7. REFERENCES

Hammond, D., Mant, J., Holloway, J., Elbourne, N. and Janes, M., 2011. Practical river restoration appraisal guidance for monitoring options (PRAGMO). The River Restoration Centre (RRC), Bedfordshire, UK. pp.26, 41-80.

Natural England, Environment Agency, Tweed Forum. 2013. River Till Restoration Strategy. [Online] Available at: http://tweedforum-org.stackstaging.com/wp-content/uploads/2018/08/TRRS_web.pdf (Accessed February 2019)

APPENDIX A

MONITORING WORK PACKAGES

(Important: The work packages provided are up-to-date as of the date of report submission. The intention is that these are working documents that may be updated by Tweed Forum at a later stage. As such, it should not be assumed that these documents are the latest version).

Wooler Water Bank Stabilisation

Site description:

The Wooler Water is a highly dynamic gravel-bed river and large flood events in 2008 and 2009 resulted in extensive channel adjustment to a section of the river immediately downstream of Wooler town. At the site in question, the river avulsed into a new location, resulting in the active erosion of a 140 m section of bank, 2-3 m in height that posed a risk to a nearby buried pipeline. Design and construction work was commissioned at the site between 2014 and 2015. The eroding bank was reprofiled to provide a more stable angle and protected along the length of its base with a matrix of natural large wood material.

Monitoring has since been undertaken, with annual (or post-flood) topographic resurveys/ analysis of the site carried out.

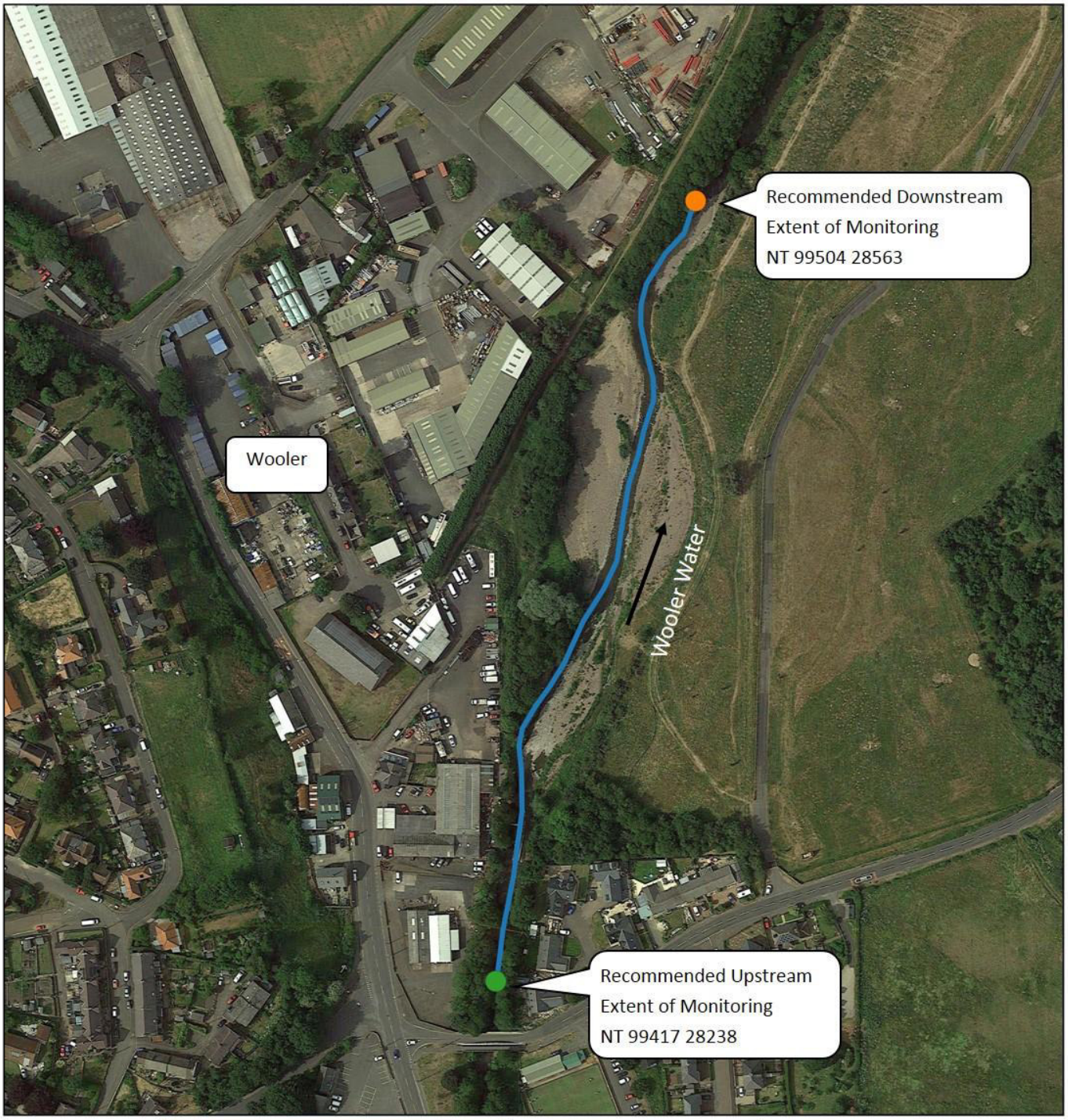
Site photos and images:





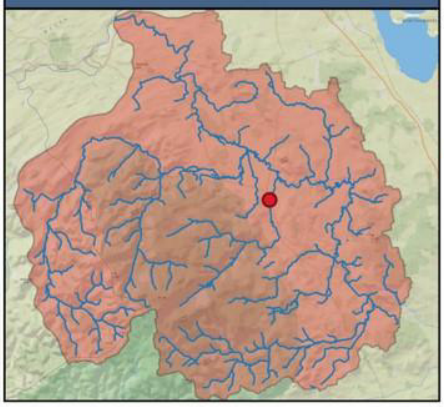
Existing Data for site:

- Photos – Pre, during and post construction (cbec)
- Photos – Pre construction (Dickinson/ Tweed Forum)
- Pre-construction topographic data (cbec)
- Pre-construction modelling (cbec)
- Drawing files (cbec)
- Project report (cbec)
- Post-construction topographic data (cbec)
- Monitoring Report (cbec)



Project Title	
Wooler Water Bank Stabilisation	
Drawing Title	
Study Extent	

Overview



Legend

- Downstream Extent
- Upstream Extent
- Channel

N

1:4000

0 50 100 m

British National Grid/GCS OSGB 1936

Map data ©2015 Google

Wooler Water Bank Stabilisation

Monitoring objectives and delivery:

Objectives

- To determine the post restoration geomorphic adjustments.
- To determine changes to sediment distribution as a result of the works.

Measurements

- Assessment of changes in the spatial distribution of alluvial bar features, channel geometry, and bed topography.
- Assessment of the changes in morphological diversity (i.e. morphological units).

Delivery

- Detailed topographical survey of the reach to monitor any changes in the spatial distribution of alluvial bar features.
- Repeated fluvial audits to provide a semi-quantitative method to assess changes in the physical condition of the channel and the spatial distribution of features such as morphological units, alluvial barforms and erosion along a continuous length of channel.

Indicative timescales/ timeline:

Detailed topographic survey: Annually.

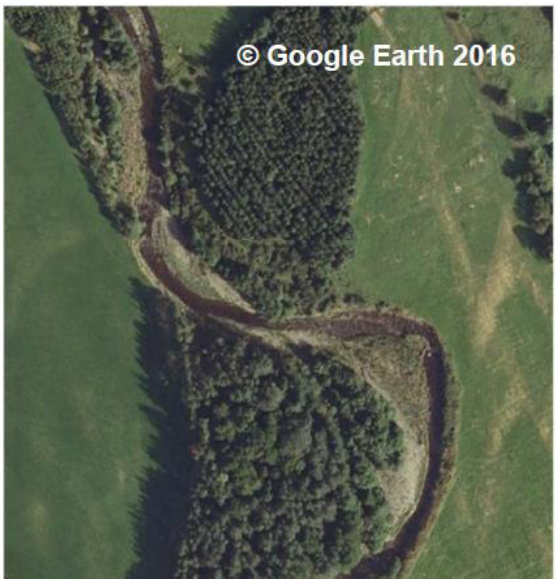
Fluvial audit: Every two years or following a high magnitude flow event (this should be done at the same time of year to ensure vegetation cover is consistent).

Haughhead Ford Removal

Project description:

The Haugh Head ford, located on the Wooler Water upstream of Wooler, Northumberland (NT 99764 24852 to NU 00250 30212), is a significant barrier to fish passage. A number of assessments have been commissioned by Tweed Forum at the site, to assess options for mitigation of the structure and the potential impacts of this. A large number of stakeholders and landowners have been involved in the discussions to-date. Sediment transport and flood risk modelling studies have been undertaken throughout the immediate reach and wider Wooler Water extents, the results being used to inform the production of designs for the site.

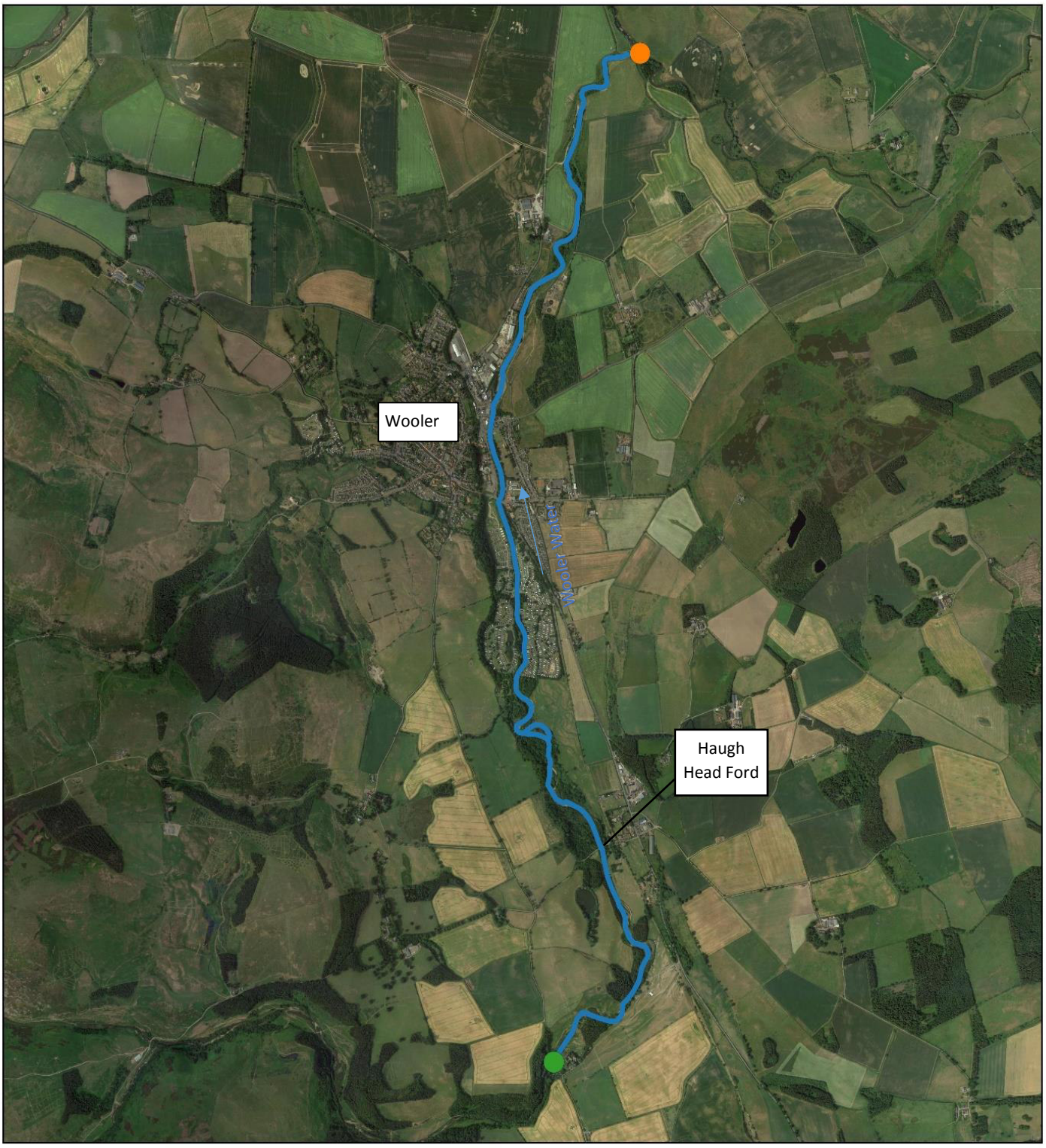
Site photos and images: Section downstream showing natural channel adjustment





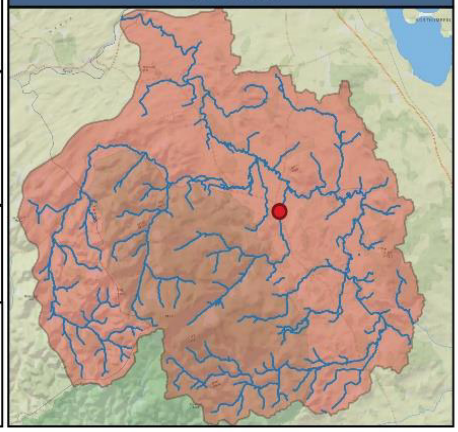
Existing Data for site:

- Wider Wooler Water Photos (cbec and Tweed Forum)
- Wider Wooler Water Drone Footage (Tweed Forum/ Rutherfords)
- Wider Wooler Water Consenting Application documentation for 'Way to Wooler' site (Tweed Forum/ Rutherfords)
- Topographic data (cbec)
- Project report (cbec)
- Flood mapping study report (JBA)
- Wooler Water Geomorphology Study (Jacobs)
- Drawing files (AECOM)
- Sediment transport modelling report (cbec)
- Still frames from timelapse footage (provided by EA)
- Ortho-rectified drone footage (EA)
- Draft design drawings (AECOM)
- Peer review (AECOM)
- Flood Model and Topographic Data (JBA/ Storm Geomatics)
- Ecological survey report – pending



Project Title	
Wooler Water Haugh Head	
Drawing Title	
Study Extent	

Overview



Legend

- Downstream Extent
- Upstream Extent
- Channel

N

1:50000

0 600 1200 m

British National Grid/GCS OSGB 1936

Map data ©2015 Google

Haughhead Ford Removal

Monitoring objectives and delivery:

Objectives

- To determine the post restoration geomorphic adjustments.
- To determine changes to sediment distribution as a result of the works.
- To determine the physical habitat diversity of the restored reach.
- The ecological response within restored and adjacent reaches, to document how the biota adjust to the changing physical habitat.

Measurements

- Assessment of the changes to sediment size distribution across the reach.
- Assessment of the changes in the physical condition of the channel (i.e. planform, presence/ absences of alluvial barforms and bank erosion).
- Assessment of the changes in morphological diversity (i.e. morphological units).
- Assessment of changes in the spatial distribution of alluvial bar features, channel geometry, and bed topography.
- Monitor presence of biota throughout the restoration site.

Delivery

- Sediment sampling at specific location.
- Repeated in-channel and floodplain ecological surveys.
- Repeated fluvial audits to provide a semi-quantitative method to assess changes in the physical condition of the channel and the spatial distribution of features such as morphological units, alluvial barforms and erosion along a continuous length of channel.
- Detailed topographical survey of the reach to monitor any changes in the spatial distribution of alluvial bar features.
- Georeferenced drone survey (with pre-programmed flight path) to assess the channel adjustment.

Indicative timescales/ timeline:

Fixed Point Photography: pre-work then annually thereafter or after high magnitude flow events, (this should be carried out at the same time of year to ensure vegetation cover is consistent).

Sediment sampling: every 2 years or following a high magnitude flow events (ideally 3 years of baseline data to monitor changes in sediments size-distribution).

Fluvial audit: every 2 years or following a high magnitude flow event (this should be done at the same time of year to ensure vegetation cover is consistent).

Detailed topographic survey: annually.

Georeferenced drone survey: every 6 months (once in winter and once in summer) ideally during low flows, recommended for at least 10 years.

Ecological survey: completed on an annual basis and seasonally to maximise species capture.

Red: Urgent. Amber: Not urgent but should be undertaken prior to works commencing. Green: Low priority or data already exists

River Glen: Kirknewton Weir Removal

Project description:

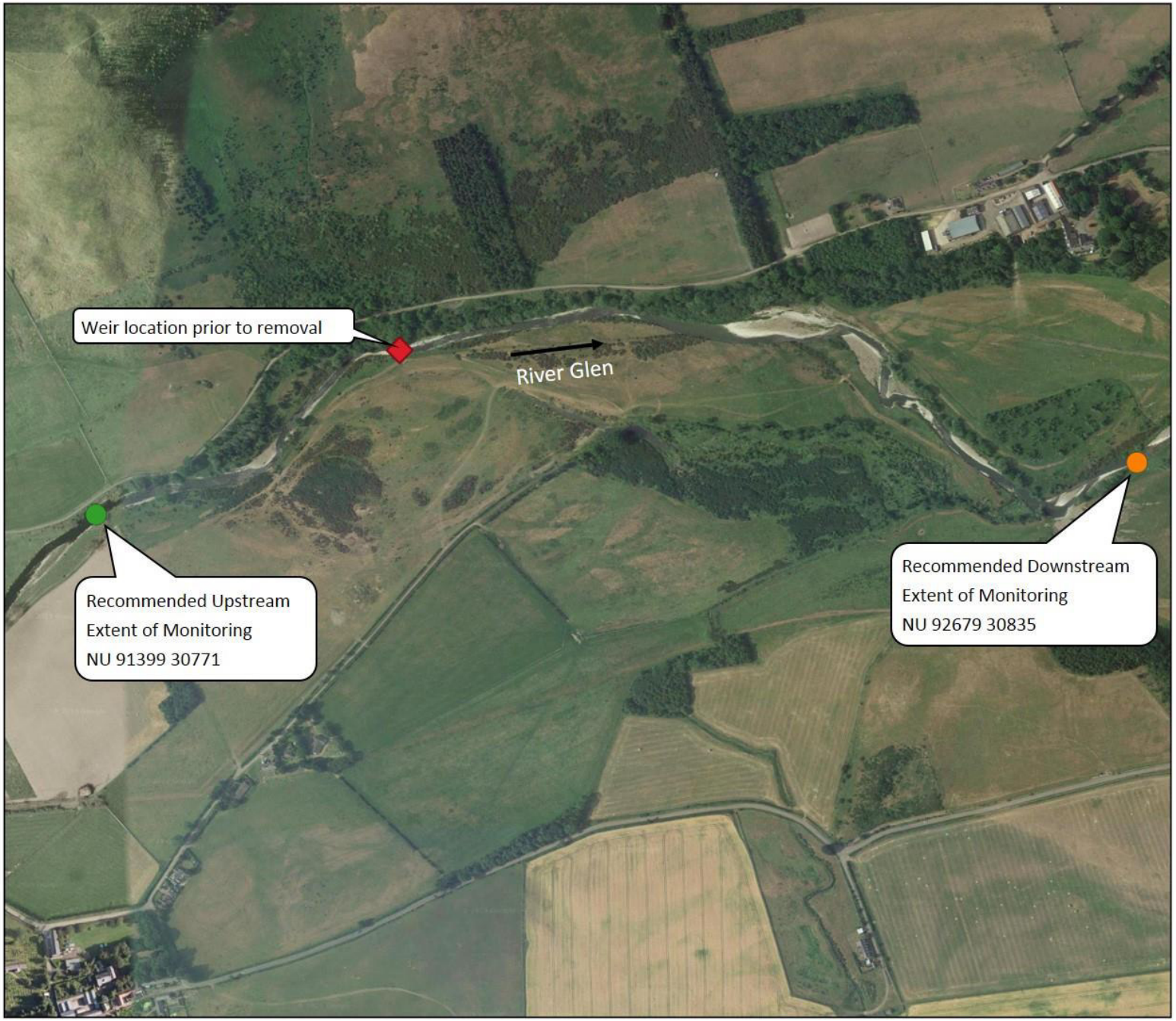
The Kirknewton weir was a defunct gauging structure, situated on the River Glen near Kirknewton, at OS NGR NT 91870 31026. Tweed Forum commissioned an options appraisal of the site in 2015, following which, outline design drawings were developed for each of the options identified. Full removal of the weir and associated structures was ultimately recommended, and construction has since been completed at the site to remove the weir.

Site photos and images:



Existing Data for site:

- Options appraisal report produced by AECOM in June 2015.
- Options development report and Appendices (including options design drawings) (Tweed Forum) produced in February 2015.
- Time-lapse video (available online at https://www.youtube.com/watch?v=JyTv4_MMV8k)
- Before and after photo of the site provided by the Environment Agency.



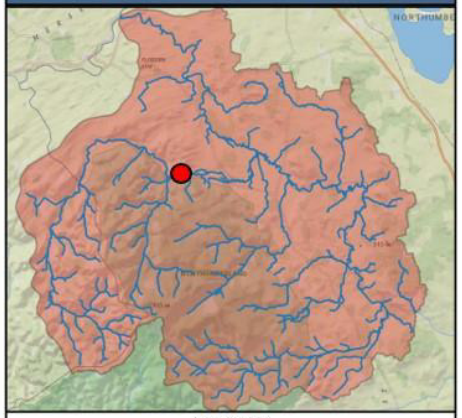
Project Title

Kirknewton Weir Removal

Drawing Title

Study Extent

Overview



Legend

- Downstream Extent
- Upstream Extent
- Weir

N

0 100 200 m

1:11500

British National Grid/GCS OSGB 1936

Map data ©2015 Google

River Glen: Kirknewton Weir Removal

Monitoring objectives and delivery:

Objectives

- To determine the post restoration geomorphic adjustments.
- To determine the physical habitat diversity of the restored reach.

Measurements

- Assessment of the changes in the physical condition of the channel (i.e. planform, presence/ absences of alluvial barforms and bank erosion).
- Assessment of the changes in morphological diversity (i.e. morphological units) within the immediate reach and subsequent upstream and downstream reaches.
- Assessment of changes in the spatial distribution of alluvial bar features, channel geometry, and bed topography.

Delivery

- Fixed point photography to provide a qualitative and visual means of monitoring geomorphic adjustment at the former weir site.
- Repeated fluvial audits to provide a semi-quantitative method to assess changes in the physical changes condition of the channel (sediment dynamics and engineering pressures) and the spatial distribution of features such as morphological units (a measure of in-channel habitat diversity), alluvial barforms and erosion along a continuous length of channel.
- Detailed topographical survey of the reach to monitor any changes in the spatial distribution of alluvial bar features.
- Georeferenced drone survey (with pre-programmed flight path) to assess the channel adjustment.

Indicative timescales/ timeline:

Fixed point photograph: Annually or after significant high magnitude flow events (this should be done at the same time of year to ensure vegetation cover is consistent).

Fluvial audit: Every 2 years or following significant flood events (this should be done at the same time of year to ensure vegetation cover is consistent).

Detailed topographic survey: Annually.

Georeferenced drone survey: Every 6 month (once in winter and once in summer) ideally during low flows.

Reedsford Embankment Breech and Planting

Project description:

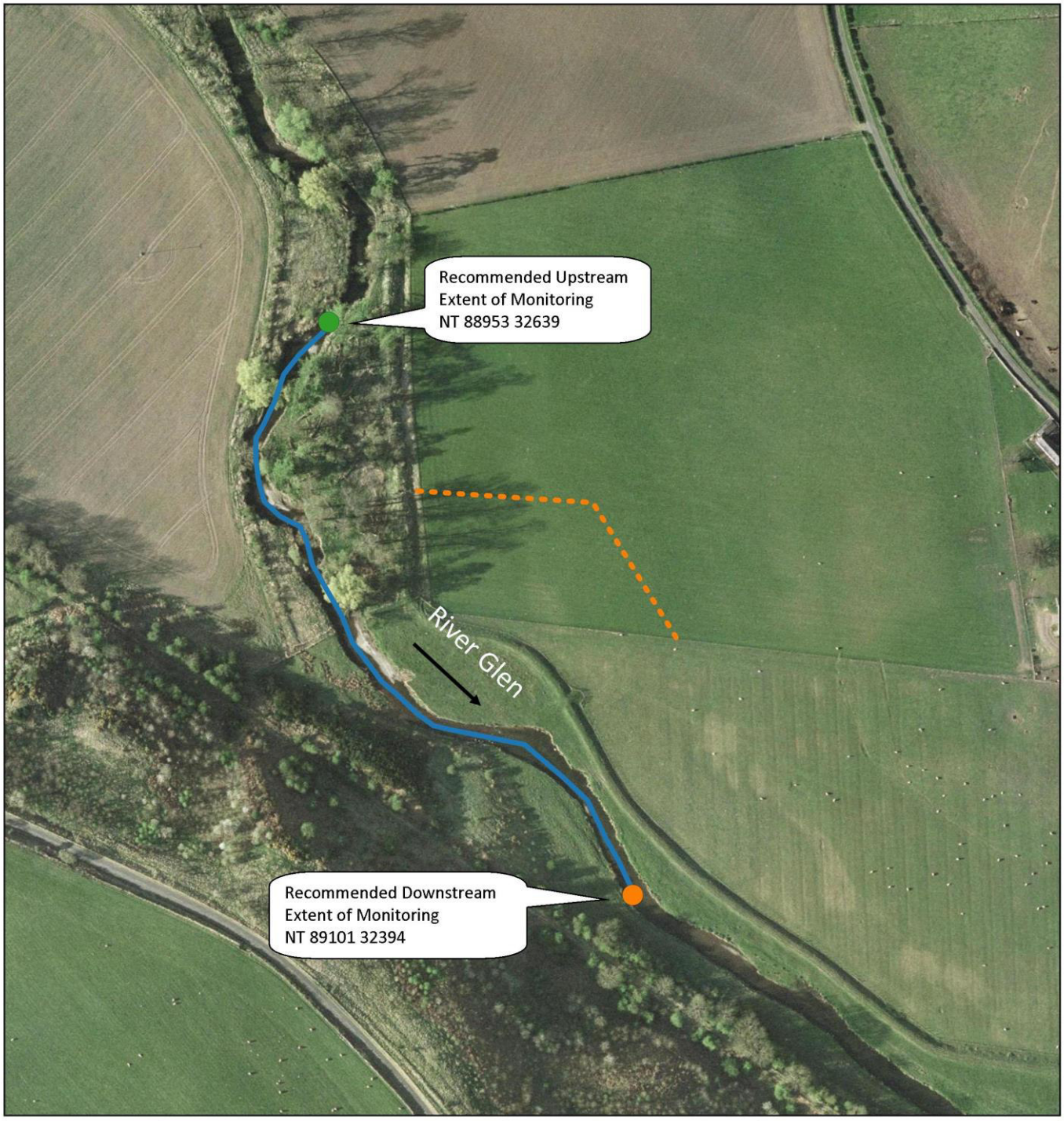
Tweed Forum have commissioned works at Reedsford, Northumberland (NT 89060 32515), with the aim of improving Natural Flood Management within the River Glen system, as well as helping to improve diversity at the site. The objective was to achieve this through implementation of a series of measures, including enlargement of an existing embankment breach (to allow greater channel – floodplain connectivity), rerouting/ erection of new fencing, and riparian planting.

Site photos and images:



Existing Data for site:

- Case study leaflet produced by the Tweed Forum and project partners which outlines the project objectives and flood bank management work.
- Site photos pre-fencing provided by Tweed Forum taken in 2014.



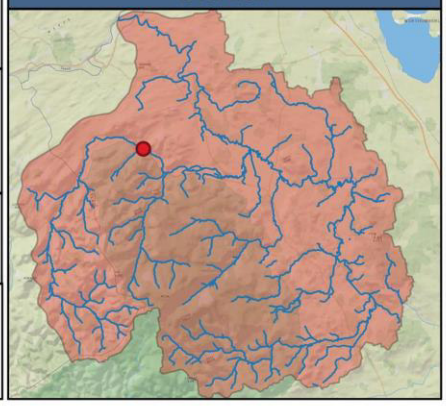
Project Title

River Glen Reedsford Embankment Breach/ Planting

Drawing Title

Study Extent

Overview



Legend

- Downstream Extent
- Upstream Extent
- New Fencing
- Channel

N

1:4000

0 50 100 m

British National Grid/GCS OSGB 1936

Map data ©2015 Google

Reedsford Embankment Breech and Planting

Monitoring objectives and delivery:

Objectives

- To determine the rate of establishment of riparian and wetland vegetation.
- To determine the ecological response within restored and adjacent reaches, to document how the biota adjust to the improved riparian and wetland habitat.

Measurements

- Assessment of the changes in the physical condition of the channel and surrounding floodplain.
- Assessment of the changes in morphological diversity (i.e. morphological units) within the reach.
- Monitor presence of biota throughout the restoration site.

Delivery

- Fixed point photography to provide a qualitative and visual means of monitoring geomorphic adjustment within the reach.
- Fluvial audit survey to monitor changes in physical habitat diversity.
- Repeated in-channel and floodplain ecological surveys.

Indicative timescales/ timeline:

Fixed point photograph: annually or after significant high magnitude flow events (this should be done at the same time of year to ensure vegetation cover is consistent).

Ecological survey: done on an annual basis and seasonally to maximise species capture.

Fluvial audit: every 2 years or following significant flood events.

River Glen: embankment set back at Lanton/ Kirknewton/ College

Project description:

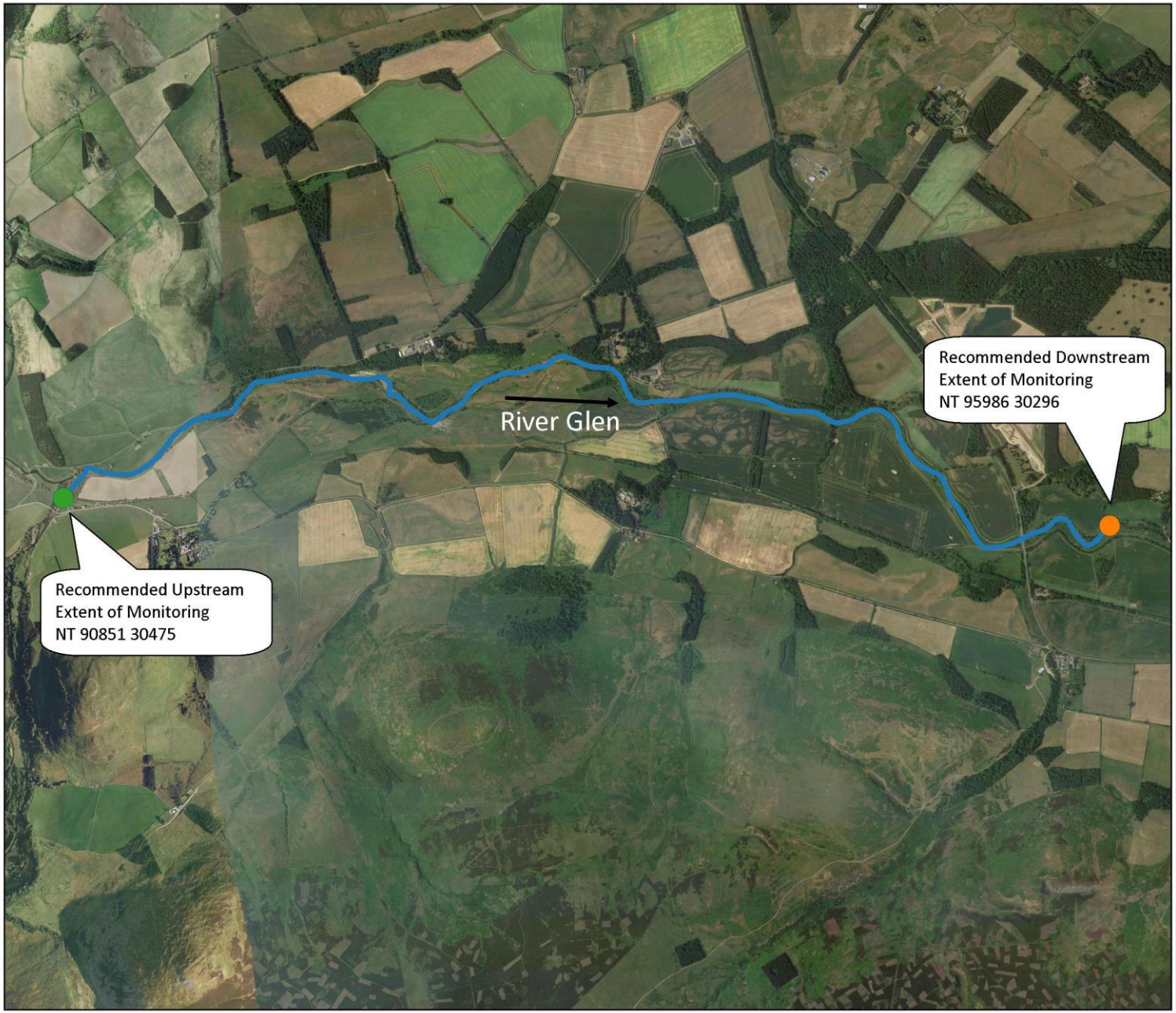
The River Glen is a high-energy gravel bed system, which is constrained within a single thread channel. The surrounding floodplain has evidence of multiple palaeo-channels illustrating the historic dynamic nature of the river. Complete or partial embankment removal has been proposed (between from College Burn through to the A697 road crossing, OS NGR NT 90851 30475 to NT 95986 30296) to reconnect the river to its surrounding floodplain. This work has involved collaboration of several landowners.

Site photos and images: Sections of channel throughout reach



Existing Data for site:

- River and floodplain management on the River Glen – Options report (AECOM)
- Ecological assessment for Lanton (EA)
- Draft design/ model report (AQUAUoS)
- Concept drawings (AQUAUoS)
- Embankment works volumes (AQUAUoS)
- Summary model results for indicative design (AQUAUoS)
- Drone footage/ imagery (Tweed Forum and EA)
- Lanton Floodbank Removal Plan (Tweed Forum)



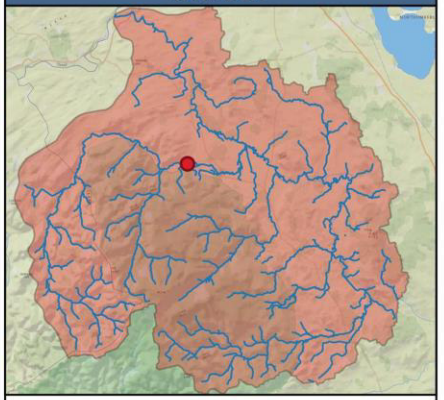
Project Title

River Glen Embankment Setback

Drawing Title

Study Extent

Overview



Legend

- Downstream Extent
- Upstream Extent
- Channel

N

0 0.5 1 km

1:45000

British National Grid/GCS OSGB 1936

Map data ©2015 Google

River Glen: embankment set back at Lanton/ Kirknewton/ College

Monitoring objectives and delivery:

Objectives

- To determine the post restoration geomorphic adjustments.
- To determine changes in hydrology and hydraulics post-works of restored and adjacent reaches, to identify the impact on flood levels.
- To determine the physical habitat diversity of the restored reach.
- The ecological response within restored and adjacent reaches, to document how the biota adjust to the changing physical habitat.

Measurements

- Assessment of the changes in the physical condition of the channel (i.e. planform, sediment transport processes).
- Assessment of the changes in the spatial distribution of alluvial bar features, channel geometry, and bed topography.
- Assessment of the changes in site hydrology.
- Monitor presence of biota throughout the restoration site.
- Assess the condition of implemented structures/ measures and effects on/ impacts to wider reach, including signs of further adjustment/ hotspots for future risk.

Delivery

- Fixed point photography to provide a qualitative and visual means of monitoring geomorphic adjustment.
- Installation of flow gauge prior to works to monitor hydrological changes within the reach (ideally need 3 year of pre-construction data).
- Repeated fluvial audits to provide a semi-quantitative method to assess changes in the physical changes condition of the channel (sediment dynamics and engineering pressures).
- Detailed topographical survey of the reach to monitor any changes bed topography and geometry.
- Repeated in-channel and floodplain ecological surveys.
- Georeferenced drone survey (with pre-programmed flight path) to assess the channel adjustment.

Indicative timescales/ timeline:

Fixed point photograph: pre-work then annually thereafter or after high magnitude flow events (this should be done at the same time of year to ensure vegetation cover is consistent).

Flow gauge: flow gauge to take readings every 15 mins, recommended for at least 10 years.

Ecological survey: done on an annual basis and seasonally to maximise species capture.

Sediment sampling: every 2 years or following a high magnitude flow events (ideally 3 years of baseline data to monitor changes in sediments size-distribution).

Fluvial audit: every 2 years or following high magnitude flood events (this should be done at the same time of year to ensure vegetation cover is consistent).

Detailed topographic survey: annually.

Georeferenced drone survey: every 6 months (once in winter and once in summer) ideally during low flows, recommended for at least 10 years.

Red: Urgent. Amber: Not urgent but should be undertaken prior to works commencing. Green: Low priority or data already exists

Lilburn Weir Removal

Project description:

The Lilburn weir structure was initially identified for removal as part of the 'assisted natural recovery' approach of the Till River Restoration Strategy, with the aim of improving in-channel natural geomorphic process, as well as benefiting fish passage.

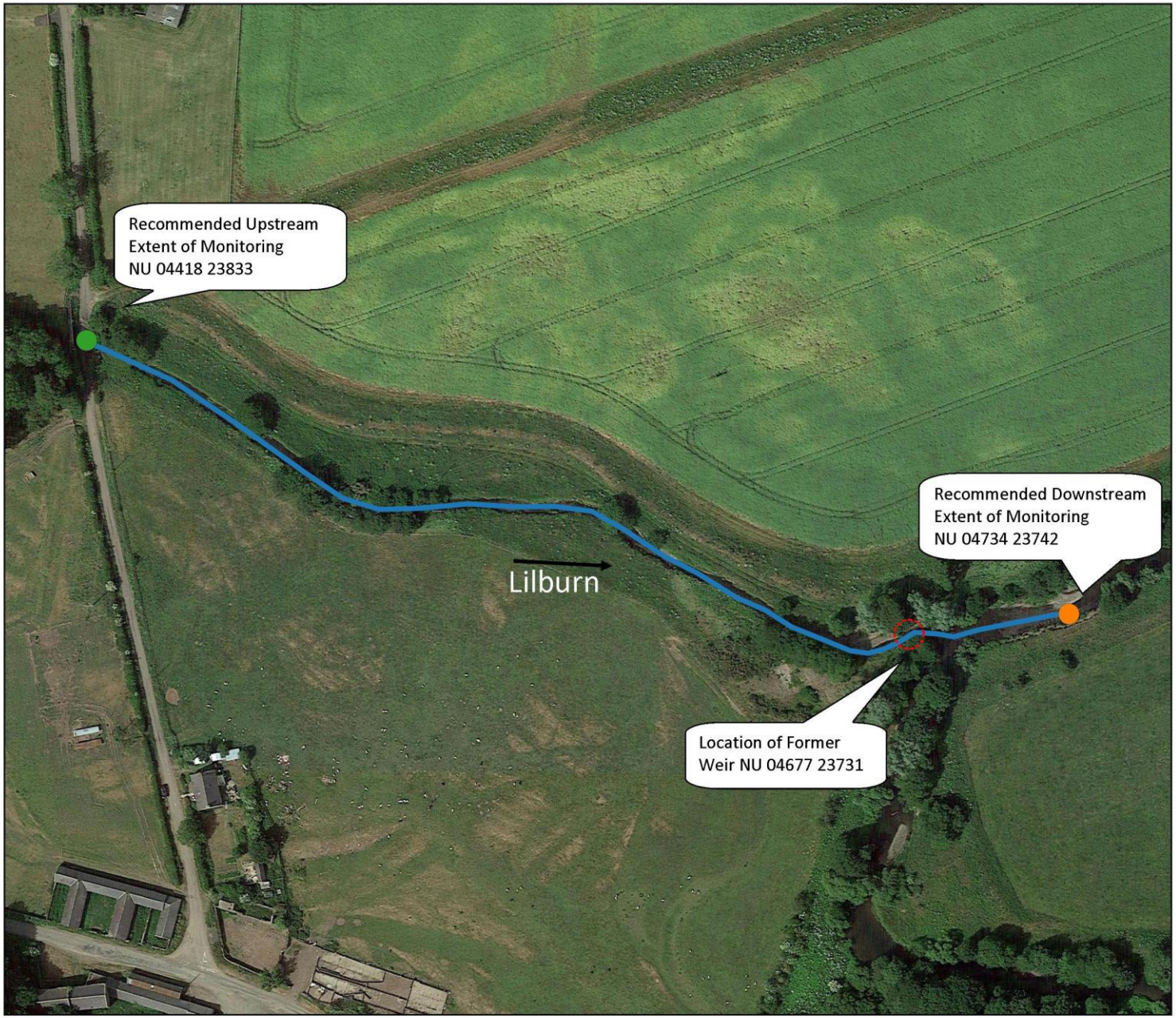
The structure, located upstream of the Lilburn confluence with the River Till at grid reference NU 4674 23735, has since been removed, with the channel and surrounding banks being reprofiled/ regraded in places.

Site photos and images: Lilburn Weir pre-removal



Existing Data for site:

- Before and after photos of the weir in pdf format.
- Pre-construction photo record of the upstream and downstream of the Lilburn weir undertaken by the Tweed Forum.
- Options appraisal report undertaken by Envirocentre in 2015. This includes a fluvial audit of 500m section of the Lilburn.



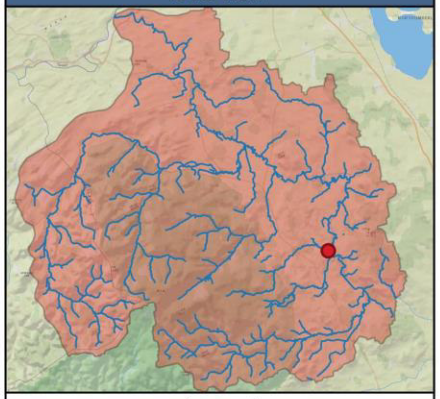
Project Title

Lilburn Weir Removal

Drawing Title

Study Extent

Overview



Legend

- Downstream Extent
- Upstream Extent
- Channel
- Location of Former Weir

N

0 25 50 m

1:3019

British National Grid/GCS OSGB 1936

Map data ©2015 Google

Lilburn Weir Removal

Monitoring objectives and delivery:

Objectives

- To determine post restoration geomorphic adjustments.
- To determine the physical habitat diversity of the restored reach.

Measurements

- Assessment of the changes in the physical condition of the channel (i.e. planform, presence/ absences of alluvial barforms and bank erosion).
- Assessment of the changes in morphological diversity (i.e. morphological units) within the immediate reach and subsequent upstream and downstream reaches.
- Assessment of changes in the spatial distribution of alluvial bar features, channel geometry, and bed topography.

Delivery

- Fixed point photography to provide a qualitative and visual means of monitoring geomorphic adjustment at the former weir site.
- Repeated fluvial audits to provide a semi-quantitative method to assess changes in the physical condition of the channel (sediment dynamics and engineering pressures) and the spatial distribution of features such as morphological units, alluvial barforms and erosion along a continuous length of channel.

Indicative timescales/ timeline:

Fixed point photograph: annually or after significant high magnitude flow events (this should be done at the same time of year to ensure vegetation cover is consistent).

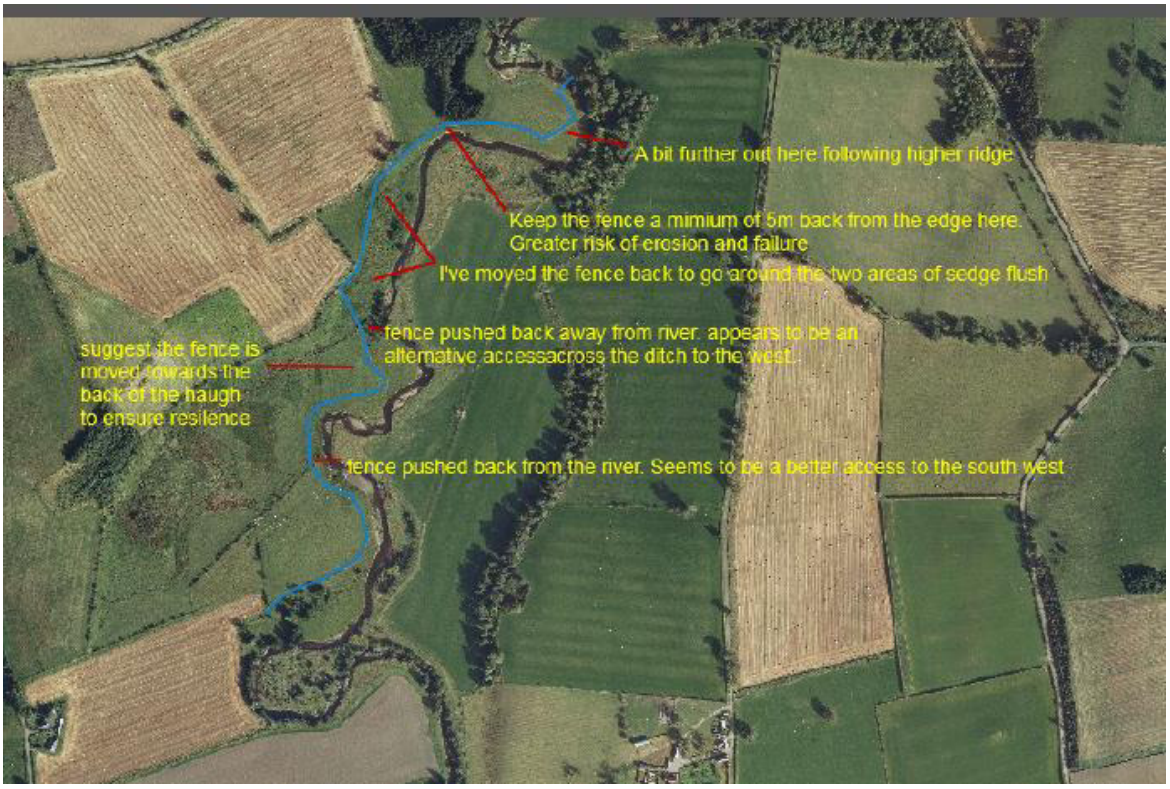
Fluvial audit: every 2 years or following high magnitude flood events (this should be done at the same time of year to ensure vegetation cover is consistent).

Lilburn Riparian Fencing

Project description:

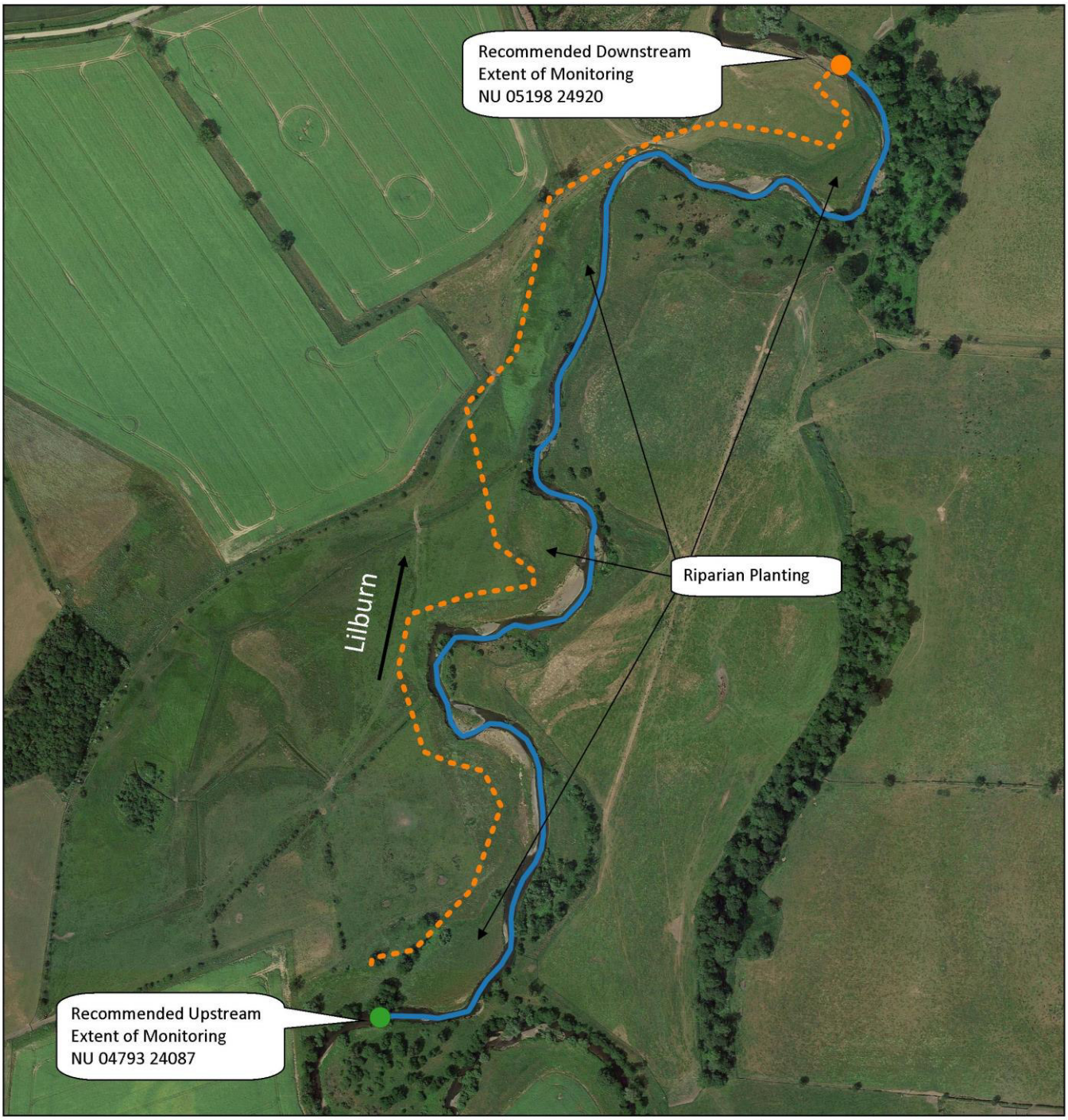
The Lilburn riparian fencing scheme focusses on a dynamic section of the Upper River Till, between Newton Mill and Newton Bridge, near Lilburn, Northumberland. The aim of the works was to install fencing along the left bank of the river to create a suitable riparian buffer zone, protected from livestock poaching. The fencing was located at such a distance so as to reduce risk of damage due to future lateral channel migration.

Site photos and images: riparian fencing plan



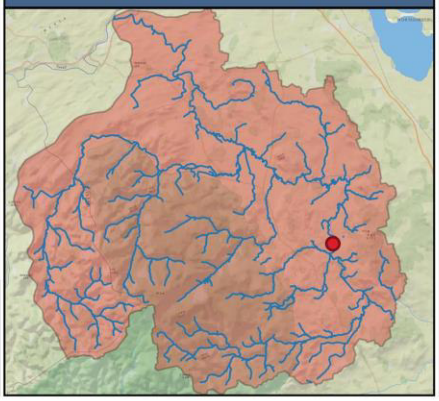
Existing data for site:

- Lilburn riparian fencing plan (Tweed Forum)
- Photos pre-fencing (Tweed Forum)



Project Title	
Lilburn Riparian Fencing	
Drawing Title	
Study Extent	

Overview



Legend

- Downstream Extent
- Upstream Extent
- Fence
- Channel

N

1:8000

0 100 200 m

British National Grid/GCS OSGB 1936

Map data ©2015 Google

Lilburn Riparian Fencing

Monitoring objectives and delivery:

Objectives

- To determine whether future adjustment at the site poses further risk to installed fencing or other adjacent areas of bank.

Measurements

- Assessment of the condition of the fencing at this location.
- Assessment of the changes in the physical condition of the channel at this location.

Delivery

- Fixed point photography to provide a qualitative and visual means of monitoring the condition of fencing and river at this location.

Indicative timescales/ timeline:

Fixed point photograph: annually or after significant high magnitude flow events (this should be done at the same time of year to ensure vegetation cover is consistent).

Fenton Riparian Fencing

Project description:

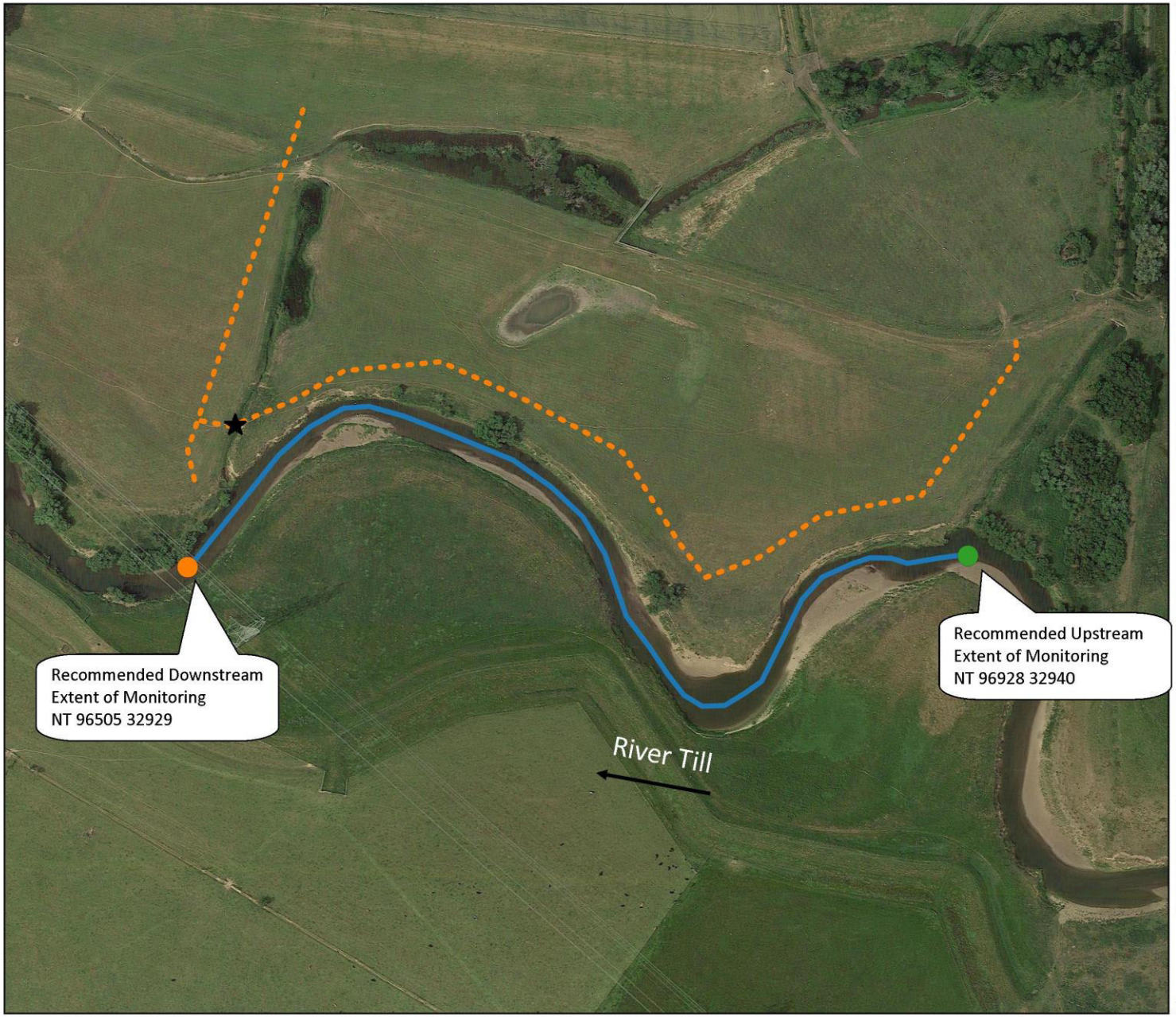
This project has involved the installation of ~850 m of temporary fencing along a river bank at West Fenton, Northumberland (NT 96928 32940 to NT 96505 32929). Riverbank erosion at the site was being exacerbated by livestock poaching. The fencing has been installed with the aim of protecting the immediate bank area from livestock, thereby encouraging vegetation to re-establish, improving stability along the bank.

Site photos and images: bank erosion pre-fencing



Existing data for site:

- Case study leaflet produced by Tweed Forum and project partners which outlines the project objectives and fencing work undertaken at Fenton.
- Plan views of fence line (Tweed Forum/ Simon Henderson).
- Maps showing historic channel adjustment (Tweed Forum/ Simon Henderson).
- Site photos pre-fencing provided by Tweed Forum taken in 2014.



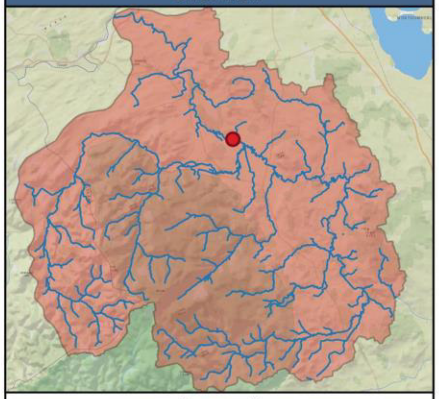
Project Title

Till Fenton Riparian Fencing

Drawing Title

Study Extent

Overview



Legend

- Downstream Extent
- Upstream Extent
- Water Gate
- New Fencing
- Channel

N

0 50 100 m

1:5036

British National Grid/GCS OSGB 1936

Map data ©2015 Google

Fenton Riparian Fencing

Monitoring objectives and delivery:

Objectives

- To determine whether future adjustment at the site poses further risk to installed fencing or other adjacent areas of bank.

Measurements

- Assessment of the condition of the fencing at this location.
- Assessment of the changes in the physical condition of the channel at this location.

Delivery

- Fixed point photography to provide a qualitative and visual means of monitoring the condition of fencing and river at this location.
- Georeferenced drone survey (with pre-programmed flight path) to assess channel adjustment.

Indicative timescales/ timeline:

Fixed point photograph: annually or after significant high magnitude flow events (this should be done at the same time of year to ensure vegetation cover is consistent).

Georeferenced drone survey: every 6 months (once in winter and once in summer) ideally during low flows.

Breamish Bank Stabilisation

Project description:

Options appraisal and outline design work has recently been undertaken at a highly dynamic reach of the River Breamish, located between Ingram and Brandon, Northumberland. The aim was to identify the most suitable restoration options to improve the physical condition of the river and protect vital infrastructure from future damage. In early 2018, following a detailed geomorphic assessment, a suite of options were identified, with two options being identified as potentially feasible. These were:

1. Sustainable 'softer' bank protection measures in four key sections of erosion.
2. In the longer term, the re-routing of the access road to Ingram, diverting it from its current alignment within the active floodplain.

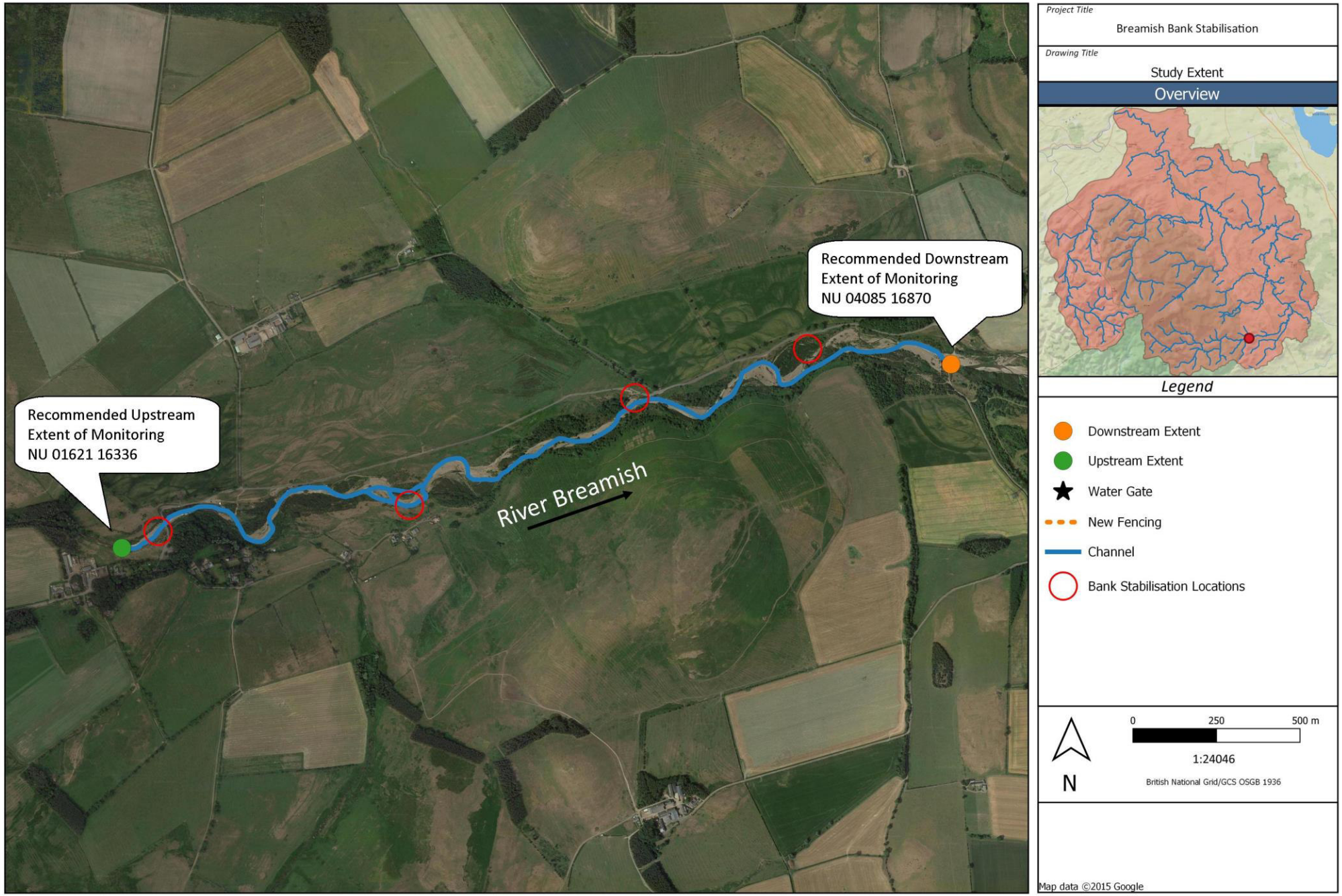
Site photos and images: Bank erosion nearing road, and existing hard bank protection.





Existing Data for site:

- Breamish options appraisal report produced by cbec in 2017, which includes a fluvial audit of the River Breamish from Ingram to Brandon Ford and Harehope Hall to Bewick Bridge.
- Maps (provided by Environment Agency).
- Photographs taken by cbec during the fluvial audit undertaken in November 2017.
- Photographs taken by Tweed Forum taken in 2014.
- Log works description outlining the proposed bank protection measures at Brandon.


Recommended Upstream
Extent of Monitoring
NU 01621 16336
Recommended Downstream
Extent of Monitoring
NU 04085 16870
River Breamish
Project Title
Breamish Bank Stabilisation
Drawing Title
Study Extent
Overview
Legend
Downstream Extent
Upstream Extent
Water Gate
New Fencing
Channel
Bank Stabilisation Locations
N
0
250
500 m
1:24046
British National Grid/GCS OSGB 1936
Map data ©2015 Google

Breamish Bank Stabilisation

Monitoring objectives and delivery:

Objectives

- To assess the stability of the implemented bank stabilisation measures.
- To monitor other areas within the reach to identify others areas where channel adjustment could pose risk to the access road.

Measurements

- Assessment of the stability and condition of implemented bank protection measures.
- Assessment of the changes in the physical condition of the channel resulting from the implementation of bank protection measures.
- Assessment of the changes in the spatial distribution of alluvial bar features, channel geometry, and bed topography to identify other areas which could be at risk of erosion.

Delivery

- Fixed point photography to provide a qualitative and visual means of monitoring geomorphic adjustment.
- Repeated fluvial audits to provide a semi-quantitative method to assess changes in the physical condition of the channel (sediment dynamics and engineering pressures) and the spatial distribution of features e.g. alluvial barforms.
- Detailed topographical survey of the reach to monitor any changes in the spatial distribution of alluvial bar features and rate of channel adjustment.
- Georeferenced drone survey (with pre-programmed flight path) to assess rate of channel adjustment.

Indicative timescales/ timeline:

Fixed point photograph: annually or after significant high magnitude flow events (this should be done at the same time of year to ensure vegetation cover is consistent).

Georeferenced drone survey: every 6 month (once in winter and once in summer) ideally during low flows.

Detailed topographic survey: every five years or after high magnitude flow events to assess rate of bank erosion

Fluvial audit: every 2 years or following high magnitude flow events (this should be done at the same time of year to ensure vegetation cover is consistent).

Red: Urgent. Amber: Not urgent but should be undertaken prior to works commencing. Green: low priority or data already exists

Breamish Restoration – Harehope Hall to Bewick Bridge

Project description:

The River Breamish, a tributary of the River Till, is a wandering gravel-bed river. Historical channel realignment and embankments have resulted in the physical condition within this reach being severely degraded. In the winter of 2017, cbec undertook a scoping study to identify potential options to improve the physical condition of the River Breamish between Harehope Hall and Bewick Bridge. Options included the setting back of embankments and channel realignment to a more natural planform.

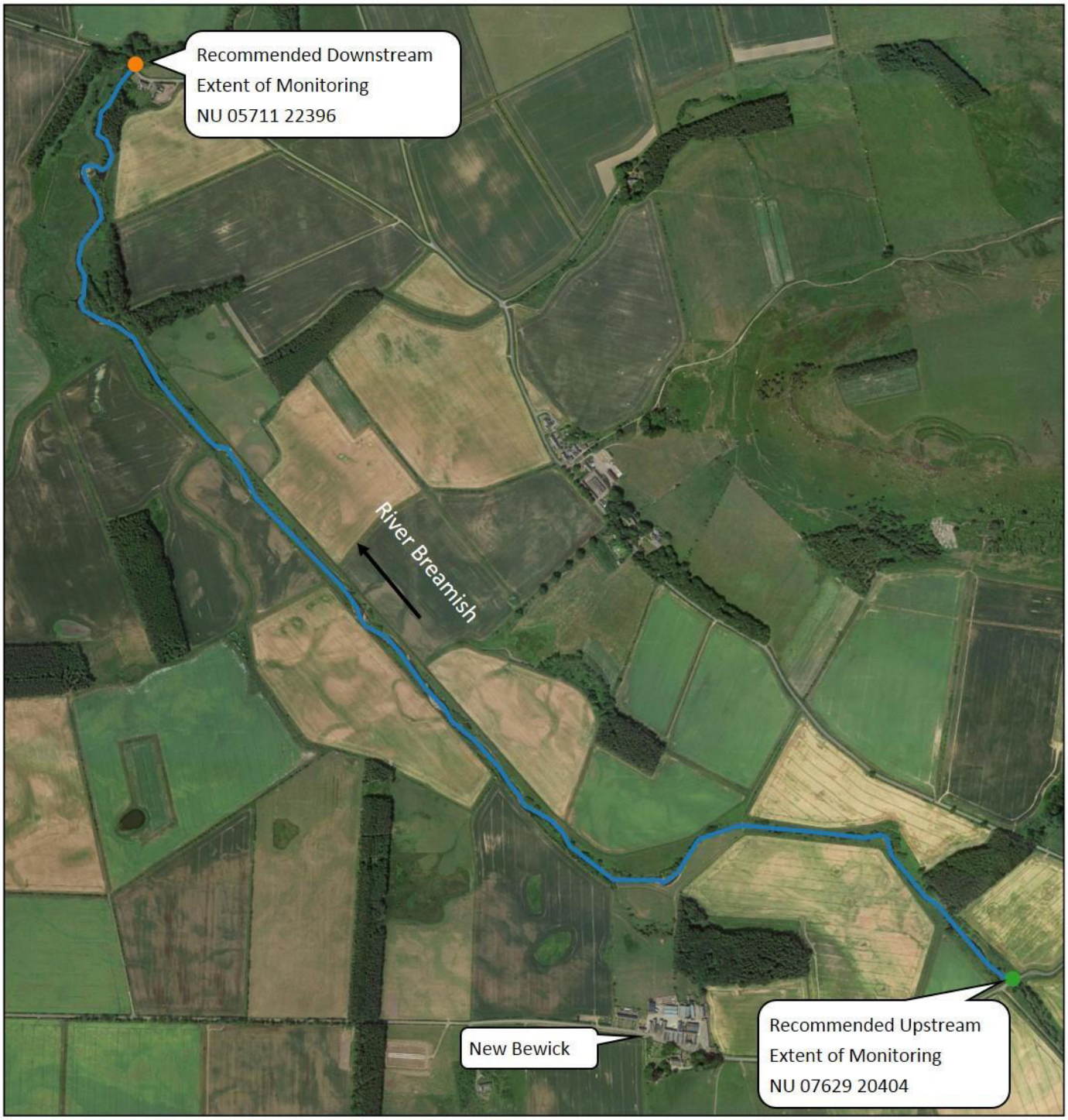
Following a period of initial landowner consultation, Tweed Forum are now looking to progress work at the site, to determine the most suitable option to take forward to detailed design.

Site photos and images: Section of straightened reach, showing lack of bedforms, and embankment (photo left)



Existing Data for site:

- Breamish Options Report produced by cbec in March 2018.
- Georeferenced photographs of the site from December 2017 (cbec)



Project Title	Overview	*Legend*	N
River Breamish - Harehope/ New Bewick Restoration		Downstream Extent	1:20000
Drawing Title		Upstream Extent	0 250 500 m
Study Extent		Channel	British National Grid/GCS OSGB 1936
			Map data ©2015 Google

Breamish Restoration – Harehope Hall to Bewick Bridge

Monitoring objectives and delivery:

Objectives

- To determine the changes in the water levels of the site.
- To determine the post restoration geomorphic adjustments.
- To determine the changes to sediment distribution as a result of the works.
- To determine the physical habitat diversity of the restored reach.
- The ecological response within restored and adjacent reaches, to document how biota adjust to the changing physical habitat.

Measurements

- Assessment of the changes in site hydrology.
- Assessment of the changes in sediment size distribution across the reach.
- Assessment of the changes to the physical condition of the channel (i.e. planform, presence/ absences of alluvial barforms and bank erosion).
- Assessment of the changes in morphological diversity (i.e. morphological units).
- Assessment of the changes in the spatial distribution of alluvial bar features, channel geometry, and bed topography.
- Monitor presence of biota throughout the restoration site.

Delivery

- Sediment sampling at specific locations.
- Repeated in-channel and floodplain ecological surveys.
- Repeated fluvial audits to provide a semi-quantitative method to assess changes in the physical condition of the channel and the spatial distribution of features such as morphological units), alluvial barforms and erosion along a continuous length of channel.
- Detailed topographical survey of the reach to monitor any changes in the spatial distribution of alluvial bar features.
- Georeferenced drone survey (with pre-programmed flight path) to assess the channel adjustment.

Indicative timescales/ timeline:

Detailed topographic survey: annually.

Flow gauge: flow gauge to take readings every 15 mins, recommended for at least 10 years.

Ecological survey: done annually on a seasonally basis to maximise species capture.

Sediment sampling: every 2 years or after high magnitude flow events.

Georeferenced drone survey: every 6 months (once in winter and once in summer) ideally during low flows, recommended for at least 10 years.

Fluvial audit: every 2 years or following high magnitude flow events (this should be done at the same time of year to ensure vegetation cover is consistent).

Red: Urgent. Amber: Not urgent but should be undertaken prior to works commencing. Green: Low priority or data already exists.



cbec

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